TRIBHUVAN UNIVERSITY — FACULTY OF MANAGEMENT

BBA Eighth Semester — Model Questions 2026 Complete Combined Solutions
Includes All Questions, Theory, Step-by-Step Numericals & Exam Memory Hints

Subjects Covered in This Combined Solution Guide:
1. **BNK 221: Corporate Financing Decision** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
2. **BNK 209: Insurance and Risk Management** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)
3. **BNK 202: Financial Derivatives** (Group A: Q1–Q10, Group B: Q11–Q17, Group C: Q18–Q21, Group D: Q22)

PAPER 1: TRIBHUVAN UNIVERSITY — BNK 221: CORPORATE FINANCING DECISION

BBA / Eighth Semester | Model Question 2026 | Full Marks: 100 | Time: 3 Hrs

Group - A: Brief Answer Questions (Attempt ALL Questions) [10 × 2 = 20]

Q1. What do you mean by corporate financing decision?

Answer: Corporate financing decision refers to the strategic financial decision made by a firm's management regarding how to raise funds to finance its long-term investments and short-term working capital needs. It determines the optimal mix of debt, preferred stock, and common equity (capital structure) that minimizes the firm's weighted average cost of capital (WACC) and maximizes the market value of the firm.

Q2. Define preemptive right.

Answer: A preemptive right is a privilege granted by a company's charter to its existing common shareholders that allows them the first option to purchase a proportionate share of any new issue of common stock (through a rights offering) before it is offered to the general public. It protects existing shareholders from dilution of ownership control and dilution of wealth.

Q3. List out the major types of leases.

Answer: The major types of leases are:
• **1. Operating Lease:** A short-term, cancellable lease where the lessor maintains and services the asset.
• **2. Financial (Capital) Lease:** A long-term, non-cancellable lease that fully amortizes the asset's cost and transfers substantially all risks and rewards of ownership to the lessee.
• **3. Sale and Leaseback:** An arrangement where a firm sells an owned asset to a financial institution and immediately leases it back for a specified period.
• **4. Leveraged Lease:** A three-party financial lease involving a lessee, a lessor (equity participant), and a third-party lender who provides the majority (60%–80%) of the financing.
• **5. Direct Lease:** A lease in which the lessee acquires the use of an asset it did not previously own directly from the manufacturer or a lessor.

Q4. Write the meaning of trust-receipt loan.

Answer: A trust-receipt loan (also known as floor-plan financing) is a secured short-term inventory loan in which the borrower holds specific, serially numbered goods (such as automobiles, heavy equipment, or consumer durables) in trust for the lender. When the borrower sells an item from the inventory, they are legally obligated to immediately remit the proceeds of that sale to the lender to repay the corresponding portion of the loan plus accrued interest.

Q5. Define financial distress.

Answer: Financial distress is a condition in which a company cannot meet, or has severe difficulty paying off, its financial obligations to its creditors as they come due. It arises from high fixed operating or financial costs (excessive debt leverage), illiquid assets, or declining operating cash flows, and if unresolved, can lead to debt restructuring, insolvency, or legal bankruptcy.

Q6. What is meant by call feature of bond?

Answer: A call feature (or call provision) in a bond indenture gives the issuing corporation the right (option) to redeem or buy back the bonds before their stated maturity date at a predetermined price (the call price, usually set above par value by a call premium). Issuers typically exercise this feature when market interest rates decline significantly below the bond's coupon rate so they can refinance at a lower interest rate.

Q7. Write about founders and angels in financing.

Answer:

• **Founders (Bootstrapping):** The entrepreneurs who establish the startup and provide the initial seed capital from their personal savings, credit, or sweat equity to validate the business idea and build an initial prototype.

• **Angel Investors:** Affluent individuals (often successful former entrepreneurs or executives) who invest their own personal funds into early-stage startups in exchange for convertible debt or ownership equity, bridging the financing gap between founders/friends-and-family and institutional venture capital.

Q8. What do you mean by moratorium period in a term loan?

Answer: A moratorium period (or grace period) in a term loan is an initial specified period granted by the lender after loan disbursement during which the borrower is not required to make principal repayments (and sometimes interest payments). It is commonly provided for project loans or construction loans so that repayment begins only after the project becomes operational and starts generating cash inflows.

Q9. Star Traders buys under terms of 3/10 net 50. Star Trader does not take discount and pays on 60th day. What is the annual percentage cost of stretching accounts payable? Assume 360 days in a year.

Answer (Step-by-Step Numerical Solution):

Given:

• Cash discount rate (d) = 3% = 0.03

• Discount period (DP) = 10 days

• Normal credit period = 50 days; Stretched payment day (CP_stretched) = 60 days

• Days in a year = 360 days

Step 1: Formula for Annual Percentage Cost (Nominal Cost) of Stretching Accounts Payable:

Annual Percentage Cost (APC) = [d / (1 - d)] × [360 / (Stretched Payment Day - Discount Period)]

Step 2: Substitute the given values:

APC = [0.03 / (1 - 0.03)] × [360 / (60 - 10)]

APC = (0.03 / 0.97) × (360 / 50)

APC = 0.0309278 × 7.2 = 0.22268 = **22.27%**

*(Note: Effective Annual Rate, EAR = (1 + 0.03/0.97)^(360/50) - 1 = (1.030928)^7.2 - 1 = **24.52%**)*

Conclusion: The annual percentage cost of stretching accounts payable is **22.27%** (or **24.52%** on an effective compounded basis).

EXAM HINT TO REMEMBER: *When accounts payable are stretched from the normal net period (day 50) to day 60, always use the ACTUAL payment day (60) minus the discount period (10) in the denominator: Days of credit used beyond discount = 60 - 10 = 50 days.*

Q10. Shikhar Company has 500,000 shares outstanding and the firm's charter provides for a cumulative voting procedure. The company has nine directors. What is the minimum number of shares needed to ensure the election of one director?

Answer (Step-by-Step Numerical Solution):

Given:

• Total number of shares outstanding (N) = 500,000 shares

• Total number of directors to be elected (D) = 9 directors

• Desired number of directors to elect (d) = 1 director

Step 1: Formula under Cumulative Voting:

Minimum Shares Required = [(N × d) / (D + 1)] + 1

Step 2: Substitute the values:

Minimum Shares Required = [(500,000 × 1) / (9 + 1)] + 1

Minimum Shares Required = (500,000 / 10) + 1 = 50,000 + 1 = **50,001 shares**

Conclusion: A minimum of **50,001 shares** is required to guarantee the election of one director under cumulative voting.

EXAM HINT TO REMEMBER: *Under Cumulative Voting, Min Shares = [(Total Shares × Desired Directors) / (Total Directors + 1)] + 1. Never forget to add '+ 1' at the end to ensure a strict majority for that seat!*

Group - B: Descriptive Answer Questions (Attempt any SIX — ALL Solved) [6 × 5 = 30]

Q11. Describe the features of optimal capital structure.

Answer: An optimal capital structure is the best proportion of debt and equity financing that minimizes a firm's overall weighted average cost of capital (WACC) and maximizes the market value of the firm (and share price). The major features of an optimal capital structure are:

- **1. Profitability (Return Maximization):** The capital structure should allow the company to utilize favorable financial leverage (trading on equity) to maximize earnings per share (EPS) and return on equity (ROE) with minimum cost.
- **2. Solvency and Liquidity (Risk Control):** It must avoid excessive debt that threatens the solvency of the firm. Fixed interest and principal repayment obligations should stay well within the firm's stable cash-flow generation capacity to avoid financial distress.
- **3. Flexibility:** The capital structure should be adaptable so the firm can easily raise additional funds or redeem costly securities when market conditions or business requirements change.
- **4. Minimum Cost of Capital:** By balancing tax-deductible cheaper debt against the rising cost of financial distress, the optimal capital structure achieves the lowest possible WACC.
- **5. Conservation of Control:** It should preserve the voting control of existing shareholders without giving excessive restrictive covenants to creditors or diluting equity unnecessarily.
- **6. Simplicity:** The financing mix should be easy to understand and manage without overly complex layers of securities that confuse investors.

Q12. Discuss the merits and demerits of long-term debt as a source of financing.

Answer: Long-term debt (such as debentures, bonds, and long-term bank loans) is a primary external source of corporate financing.

Merits (Advantages) of Long-Term Debt:
• **1. Tax Shield Benefit:** Interest paid on debt is tax-deductible, which lowers the effective after-tax cost of debt: $k_{dt} = k_d(1 - T)$.
• **2. No Dilution of Ownership Control:** Bondholders and term-loan lenders do not have voting rights, so existing common shareholders retain full control of the company.
• **3. Financial Leverage (Trading on Equity):** When the firm's return on assets (ROA) exceeds the interest rate on debt, fixed-cost debt magnifies the Earnings Per Share (EPS) and Return on Equity (ROE) for shareholders.
• **4. Fixed Claim on Income:** Creditors receive only a fixed interest rate regardless of how high the company's profits grow during prosperous years.
• **5. Callability / Flexibility:** Debt can include a call provision allowing the firm to retire the bonds early if interest rates fall.

Demerits (Disadvantages) of Long-Term Debt:
• **1. Financial Risk and Bankruptcy Threat:** Interest and principal repayments are legally binding fixed obligations; failure to pay can force the company into insolvency and liquidation.
• **2. Restrictive Covenants:** Debt indentures often impose strict limits on dividend payments, working capital ratios, further borrowing, and asset sales, reducing managerial flexibility.
• **3. Fixed Maturity and Sinking Fund Burden:** Unlike common stock, debt has a finite maturity and requires periodic sinking-fund cash outflows or refinancing.
• **4. Collateral Requirement:** Secured long-term debt requires pledging fixed assets, reducing the firm's future borrowing capacity during emergencies.

Q13. Explain the concept and importance of venture capital financing.

Answer:
Concept of Venture Capital Financing:
Venture capital (VC) is a form of private equity financing provided by institutional venture capital firms or funds to high-growth-potential, early-stage, and innovative startup companies. Because such young companies lack tangible collateral and stable cash flows to obtain bank loans, venture capitalists invest long-term equity or quasi-equity (convertible preferred stock or convertible debentures) and actively participate in mentoring, strategic planning, and corporate governance until exiting via an IPO or acquisition.

Importance of Venture Capital Financing:

• **1. Financing Innovation and High-Risk Startups:** VC bridges the capital gap for technology, biotech, and innovative ventures that traditional commercial banks reject due to high uncertainty.

• **2. Managerial and Strategic Expertise:** Beyond capital, VCs provide experienced board guidance, industry networking, talent recruitment, and financial discipline.

• **3. Economic Growth and Job Creation:** Venture-backed firms commercialize new technologies, boost productivity, and generate skilled employment.

• **4. Credibility and Follow-On Funding:** Backing from a reputable VC firm signals quality to suppliers, customers, investment bankers, and future institutional investors.

• **5. No Immediate Debt Service Burden:** Since VC is provided as equity or convertible preferred stock, startups do not face fixed monthly interest payments during their cash-burn growth phase.

Q14. What is leveraged buyout? Describe the characteristics of successful LBO.

Answer:

Meaning of Leveraged Buyout (LBO):

A Leveraged Buyout (LBO) is the acquisition of a public or private company (or a division of a company) by a small group of investors (typically a private equity sponsor and/or existing management, known as a Management Buyout or MBO) financed predominantly with borrowed funds (often 60% to 85% debt). The target company's own assets serve as collateral for the borrowed money, and its future operating cash flows are used to service and pay down the heavy debt load.

Characteristics of a Successful LBO Candidate:

• **1. Stable and Predictable Cash Flows:** Because the firm must meet heavy interest and principal schedules, a mature business with steady, non-cyclical operating cash flows is essential.

• **2. Strong, Experienced Management Team:** Capable managers with direct equity ownership incentives align their interests with debt reduction and operational efficiency.

• **3. Substantial Tangible Assets (Collateral Base):** High fixed assets and working capital allow the acquirer to secure lower-cost senior bank debt.

• **4. Low Existing Debt (Unused Debt Capacity):** A target with a clean balance sheet provides ample room to add new leverage.

• **5. Cost-Cutting and Efficiency Potential:** Opportunities to eliminate wasteful corporate overhead, improve working capital, and boost EBITDA margins.

• **6. Divestible Non-Core Assets:** Underperforming or non-core divisions that can be sold off immediately after the buyout to pay down a chunk of the acquisition debt.

Q15. Differentiate between operating lease and financial lease.

Answer: The key differences between an Operating Lease and a Financial (Capital) Lease are presented below:

Basis of Difference	Operating Lease	Financial (Capital) Lease
1. Lease Duration	Short-term; significantly shorter than the useful economic life of the asset.	Long-term; covers the major part (usually 75% or more) of the asset's economic life.
2. Revocability (Cancellation)	Cancellable by the lessee upon giving short notice.	Non-cancellable during the primary lease period without heavy penalty.
3. Maintenance & Insurance	Borne by the lessor (owner).	Borne by the lessee (user).
4. Obsolescence Risk	Rests with the lessor.	Transferred to the lessee.
5. Amortization of Cost	Single lease contract does not fully recover the asset's cost (not fully amortized).	Lease payments fully amortize the lessor's capital cost plus a return.
6. Purchase Option at End	Usually no bargain purchase option at maturity.	Often includes an option to buy the asset at a nominal or residual value.

Q16. The Mega Company has warrants outstanding that expire in three years. Each warrant entitles the holder to purchase two shares of common stock at an exercise price of Rs 150 per share. Determine theoretical value and premium over the theoretical value if the respective prices of common stock and warrants are: a. Rs 140 per share and Rs 10 per warrant. b. Rs 160 per share and Rs 40 per warrant.

Answer (Step-by-Step Numerical Solution):
Given:
• Number of shares per warrant (N) = 2 shares
• Exercise price per share (E) = Rs 150
• Formula for Theoretical Value of a Warrant (TV):
TV = max[(P_0 - E) × N , 0] *(Note: Formula value is (P_0 - E) × N, and minimum theoretical value cannot be negative, i.e., Rs 0).*
• Formula for Warrant Premium = Market Price of Warrant (W_m) - Theoretical Value (TV)

Case (a): When Stock Price (P_0) = Rs 140 and Warrant Market Price (W_m) = Rs 10:
• **Step 1: Theoretical Value (TV):**
Formula value = (P_0 - E) × N = (140 - 150) × 2 = -10 × 2 = **-Rs 20**
Since a warrant is an option and need not be exercised when out-of-the-money, its minimum floor Theoretical Value = **Rs 0**.
• **Step 2: Premium over Theoretical Value:**
Using minimum floor TV = Rs 0:
Premium = W_m - TV = Rs 10 - Rs 0 = **Rs 10**
*(Alternative convention using unfloored formula TV = -Rs 20: Premium = 10 - (-20) = **Rs 30**. Both are shown for completeness.)*

Case (b): When Stock Price (P_0) = Rs 160 and Warrant Market Price (W_m) = Rs 40:
• **Step 1: Theoretical Value (TV):**
TV = (P_0 - E) × N = (160 - 150) × 2 = 10 × 2 = **Rs 20**
• **Step 2: Premium over Theoretical Value:**
Premium = W_m - TV = Rs 40 - Rs 20 = **Rs 20**

EXAM HINT TO REMEMBER: *TV = (Market Price of Stock - Exercise Price) × Number of shares per warrant. When P_0 < E, formula value is negative (-Rs 20), so effective TV is Rs 0. Write BOTH the floor TV (Rs 0 -> Premium Rs 10) and formula TV (-Rs 20 -> Premium Rs 30) in the exam to guarantee full marks regardless of the examiner's key!*

Q17. Akash Food Company (AFC) projects an increase in sales from Rs 1,200,000 to Rs 1,800,000 but it needs an additional Rs 200,000 of current assets to support this expansion. The money can be obtained from the bank at an interest rate of 12 percent, discount interest; no compensating balance is required. Alternatively, AFC can finance the expansion by no longer taking discounts, thus increasing accounts payable. AFC purchases on the terms of 3/10, net 40, but it can delay payment for an additional 20 days and thus becoming 20 days past due without a penalty. Based strictly on effective annual interest rate comparisons, how should AFC finance its expansion?

Answer (Step-by-Step Numerical Solution):
Given:
• Additional financing needed = Rs 200,000
• **Option 1 (Bank Loan):** Discount interest loan at stated rate (i) = 12% = 0.12, no compensating balance.
• **Option 2 (Stretching Trade Credit):** Terms 3/10, net 40, delayed by an extra 20 days -> Actual payment day = 40 + 20 = 60th day.

Step 1: Calculate Effective Annual Rate (EAR) of the Discount Bank Loan:
EAR (Bank Loan) = i / (1 - i)
EAR (Bank Loan) = 0.12 / (1 - 0.12) = 0.12 / 0.88 = 0.136364 = **13.64%**

Step 2: Calculate Cost of Stretching Trade Credit (3/10, net 60):
• **Annual Percentage Cost (Nominal Rate, assuming 360-day year):**
APC = [d / (1 - d)] × [360 / (CP_stretched - DP)]
APC = [0.03 / (1 - 0.03)] × [360 / (60 - 10)] = (0.03 / 0.97) × (360 / 50) = 0.030928 × 7.2 = **22.27%**
*(Using 365 days: APC = (0.03 / 0.97) × (365 / 50) = **22.58%**)*
• **Effective Annual Rate (Compounded EAR):**
EAR (Trade Credit, 360 days) = (1 + 0.03 / 0.97)^(360 / 50) - 1 = (1.030928)^7.2 - 1 = **24.52%**
*(Using 365 days: EAR = (1.030928)^7.3 - 1 = **24.90%**)*

Step 3: Decision / Recommendation:
Since the effective annual cost of the discount bank loan (**13.64%**) is substantially lower than the cost of forgoing discounts and stretching accounts payable (**22.27% nominal / 24.52% EAR**), **AFC should finance its expansion by**

borrowing from the bank and continue taking the 3% cash discount on its purchases.

EXAM HINT TO REMEMBER: *Discount loan EAR = Interest / (1 - Interest). Trade credit paid 20 days late on 3/10 net 40 means actual payment day is 40 + 20 = 60. Compare 13.64% (Bank) vs 22.27% / 24.52% (Trade Credit) and choose the lower cost!*

Group - C: Analytical Answer Questions (Attempt any THREE — ALL Solved) [3 × 10 = 30]

Q18. Explain the methods of issuing securities. Also discuss the process of initial public offering.

Answer:

Part 1: Methods of Issuing Securities

Corporations raise long-term capital in the primary market through the following major methods:

• **1. Public Issue through Prospectus (Initial Public Offering & Follow-on Public Offering):**

- *Initial Public Offering (IPO):* When an unlisted private company sells its shares to the general public for the first time.

- *Follow-on Public Offering (FPO) / Seasoned Equity Offering:* When an already listed public company issues additional shares to the general public.

• **2. Rights Offering (Privileged Subscription):** Issuing new shares directly to existing shareholders in proportion to their current shareholding at a discounted subscription price (preserving their preemptive right).

• **3. Private Placement:** Selling securities directly to a small number of sophisticated institutional investors (such as pension funds, insurance companies, mutual funds, or venture capital firms) without a public prospectus. It is faster and has lower flotation costs.

• **4. Offer for Sale:** The issuing company sells the entire block of securities to an issue house or investment bank at an agreed price, which then resells the securities to the public.

• **5. Shelf Registration / Book Building / Dutch Auction:** Modern pricing and issuance mechanisms where institutional bids determine the cut-off price (book building) or pre-registered securities are sold gradually over time (shelf registration).

Part 2: Process of Initial Public Offering (IPO)

• **Step 1: Board & Shareholder Approval:** The Board of Directors and Annual/Special General Meeting pass a resolution approving the capital increase and public issue.

• **Step 2: Appointment of Issue Manager (Underwriter) & Due Diligence:** The company appoints a licensed merchant/investment banker to act as Issue Manager, Underwriter, and Registrar to the Issue.

• **Step 3: Credit Rating & Valuation:** Obtain a credit rating from an accredited rating agency (when required by regulations) and prepare audited financial statements and a draft prospectus.

• **Step 4: Regulatory Filing and Approval (SEBON / SEC):** Submit the registration statement and draft prospectus to the securities regulator (e.g., SEBON in Nepal) for review and approval during the cooling-off/waiting period.

• **Step 5: Publication of Prospectus & Issue Opening:** Once approved, publish the prospectus/announcement in national newspapers and open applications to the public (e.g., via ASBA/C-ASBA system).

• **Step 6: Subscription Closure, Allotment & Refund:** Close the issue, verify applications, allot shares proportionately or by lottery (as per securities allotment guidelines), and refund unallotted funds.

• **Step 7: Demat Credit & Stock Exchange Listing:** Credit allotted shares into investors' Demat accounts (CDSC) and list the shares on the stock exchange (e.g., NEPSE) for secondary market trading.

Q19. The Nepal Energy Company has announced a rights offer to raise Rs 600 million for a new Gas Station. The stock currently sells for Rs 600 per share, and there are 4.8 million shares outstanding. Par value is Rs 100. a. What is the maximum possible subscription price? What is the minimum? b. If the subscription price is set at Rs 500 per share, how many shares must be sold? c. How many rights will it take to buy one share? d. What is the value of a right? e. What is the ex-rights price? f. Show how a shareholder with 1,000 shares before the offering and no desire (or money) to buy additional shares is not harmed by the rights offer.

Answer (Step-by-Step Numerical Solution):

Given:

• Funds to be raised = Rs 600,000,000 (Rs 600 million)

• Rights-on market price of stock (P_0) = Rs 600 per share

• Existing shares outstanding (N_0) = 4,800,000 shares (4.8 million)

• Par value per share = Rs 100

a. Maximum and Minimum Possible Subscription Price:

• **Maximum Subscription Price:** Theoretically, the subscription price cannot exceed the current market price of the stock, i.e., **Rs 600 per share** (otherwise rational investors would buy shares directly in the market rather than exercising rights).

• **Minimum Subscription Price:** Legally, a company cannot issue shares at a discount below par value (or technically >

Rs 0 in jurisdictions without par value restrictions, and in Nepal Corporate Law, minimum is the par value = **Rs 100 per share**).

b. Number of New Shares to be Sold (if Subscription Price, P_s = Rs 500):
New Shares (N_1) = Funds to be Raised / Subscription Price (P_s)
N_1 = Rs 600,000,000 / Rs 500 = **1,200,000 shares (or 1.2 million shares)**

c. Number of Rights Required to Buy One New Share (#):
Number of Rights per New Share (#) = Existing Shares (N_0) / New Shares (N_1)
= 4,800,000 / 1,200,000 = **4 rights**

d. Theoretical Value of One Right (V_r):
V_r = (P_0 - P_s) / (# + 1)
V_r = (600 - 500) / (4 + 1) = 100 / 5 = **Rs 20 per right**

e. Ex-Rights Price of Stock (P_e):
P_e = P_0 - V_r = Rs 600 - Rs 20 = **Rs 580 per share**
(Verification: P_e = [(N_0 × P_0) + (N_1 × P_s)] / (N_0 + N_1) = [(4.8M × 600) + (1.2M × 500)] / 6.0M = 3,480M / 6.0M = Rs 580)

f. Proof that a Shareholder with 1,000 Shares Who Does Not Exercise Rights is Not Harmed (by Selling Rights):
• **Wealth BEFORE Rights Offering:**
Value of 1,000 shares @ Rs 600 = 1,000 × Rs 600 = **Rs 600,000**
• **Wealth AFTER Rights Offering (Selling all 1,000 Rights):**
- Value of 1,000 shares at Ex-Rights Price (Rs 580) = 1,000 × Rs 580 = Rs 580,000
- Cash received from selling 1,000 rights @ Rs 20 each = 1,000 × Rs 20 = Rs 20,000
- **Total Wealth After Offering** = Rs 580,000 + Rs 20,000 = **Rs 600,000**
Since total wealth remains unchanged at **Rs 600,000**, the shareholder is not harmed as long as they sell their rights rather than letting them expire worthless.

EXAM HINT TO REMEMBER: *Rights Offering 4-step chain: (1) New Shares = Funds / P_s; (2) Rights per new share (#) = Old Shares / New Shares; (3) Value of 1 Right (V_r) = (P_0 - P_s) / (# + 1); (4) Ex-rights price (P_e) = P_0 - V_r = P_s + (# × V_r).*

Q20. The Kathmandu Electronic Company (KEC) intends to borrow Rs 800,000 to support its short-term financing requirements during the next year. The financing alternatives offered by the bank include: Alternative 1: A discount interest loan with a simple interest of 15 percent and no compensating balance requirement. Alternative 2: A 14 percent simple interest loan that has a 15 percent compensating balance requirement. Alternative 3: Rs 1 million revolving line of credit with simple interest of 12 percent paid on the amount borrowed and a 1 percent commitment fee. a. Compute the effective cost (rate) of each financing alternative assuming KEC borrows Rs 800,000. Which alternative should it use? b. Discuss the factors other than cost to be considered while choosing a bank for short-term loan.

Answer (Step-by-Step Numerical Solution):
Given: Amount borrowed / usable funds required = Rs 800,000; Total credit line in Alt 3 = Rs 1,000,000.

Part (a): Effective Cost of Each Financing Alternative:
• **Alternative 1: 15% Discount Interest Loan (No Compensating Balance):**
Effective Cost = Interest Rate / (1 - Interest Rate) = 0.15 / (1 - 0.15) = 0.15 / 0.85 = 0.176471 = **17.65%**

• **Alternative 2: 14% Simple Interest Loan with 15% Compensating Balance:**
Effective Cost = Interest Rate / (1 - Compensating Balance Ratio) = 0.14 / (1 - 0.15) = 0.14 / 0.85 = 0.164706 = **16.47%**

• **Alternative 3: Rs 1,000,000 Revolving Line of Credit at 12% Interest + 1% Commitment Fee:**
- Interest on Rs 800,000 borrowed @ 12% = Rs 800,000 × 0.12 = Rs 96,000
- Commitment fee on unused credit (Rs 1,000,000 - Rs 800,000 = Rs 200,000) @ 1% = Rs 200,000 × 0.01 = Rs 2,000
- Effective Cost (Commitment Fee on Unused Portion) = (Rs 96,000 + Rs 2,000) / Rs 800,000 = Rs 98,000 / Rs 800,000 = **12.25%**

*(Note: Even if the 1% commitment fee applies to the total Rs 1,000,000 line = Rs 10,000, Effective Cost = (96,000 + 10,000) / 800,000 = **13.25%**.)*

• **Decision:** KEC should choose **Alternative 3 (Revolving Line of Credit)** because it has the lowest effective annual cost (**12.25%**, or 13.25%).

Part (b): Factors Other Than Cost When Choosing a Bank for a Short-Term Loan:
• **1. Reliability and Willingness to Assume Risk:** The bank's policies toward supporting clients during temporary industry downturns (loyalty during tight credit periods).
• **2. Maximum Loan Size (Lending Limit):** Whether the bank has sufficient capital and legal lending limits to meet the firm's growing borrowing needs.
• **3. Advisory and Ancillary Services:** Quality of cash management, letter of credit (LC), foreign exchange, and financial counseling services offered by the bank.
• **4. Specialization and Industry Experience:** Familiarity of the bank's loan officers with the borrower's specific industry and working capital cycle.
• **5. Flexibility of Loan Covenants and Collateral Terms:** Reasonableness of security requirements, compensating balances, and repayment schedules.

EXAM HINT TO REMEMBER: *Remember the 3 bank loan denominators: (1) Discount loan subtracts 'i' in denominator -> i/(1-i); (2) Simple loan with CB subtracts 'CB' -> i/(1-CB); (3) Discount loan WITH CB subtracts BOTH -> i/(1-i-CB). For Revolving Credit, add Interest + Commitment Fee in the numerator and divide by usable borrowed amount!*

Q21. Himalaya Company can lease equipment for three years, making annual payments of Rs 150,000 per year at the end of each year or they can buy the equipment for Rs 300,000. At the end of third years, the equipment will have Rs 60,000 salvage value. The firm's before tax cost of debt is 10 percent. The company uses straight-line depreciation and has a 40 percent tax rate. a. Calculate cost of leasing. b. Calculate cost of purchasing. Should the machine be leased or purchased? c. The appropriate discount rate for cash flows used in the analysis is the firm's after-tax cost of debt. Why?

Answer (Step-by-Step Numerical Solution):
Given:
• Lease payment (L) = Rs 150,000 per year (at the end of Years 1, 2, and 3)
• Purchase price of equipment (I_0) = Rs 300,000; Life (n) = 3 years; Salvage value (SV) = Rs 60,000
• Before-tax cost of debt (k_d) = 10%; Corporate tax rate (T) = 40%
• **After-tax cost of debt (k_dt)** = k_d × (1 - T) = 10% × (1 - 0.40) = **6%**
• Present Value factors at 6% for 3 years:
- PVIFA(6%, 3 yrs) = [1 - (1.06)^(-3)] / 0.06 = **2.6730**
- PVIF(6%, 3 yrs) = (1.06)^(-3) = **0.8396**

a. Cost of Leasing (Present Value of After-Tax Lease Outflows):
• Annual after-tax lease payment = L × (1 - T) = Rs 150,000 × (1 - 0.40) = Rs 90,000 per year
• PV of Cost of Leasing = Rs 90,000 × PVIFA(6%, 3 yrs)
= Rs 90,000 × 2.673012 = **Rs 240,571.08** (or Rs 240,570 using 2.6730)

b. Cost of Purchasing (Borrow & Buy) and Decision:
• Annual Straight-Line Depreciation = (Cost - Salvage Value) / Life = (300,000 - 60,000) / 3 = Rs 80,000 per year
• Annual Depreciation Tax Shield = Depreciation × T = Rs 80,000 × 0.40 = Rs 32,000 per year
• After-tax Salvage Value in Year 3 = Rs 60,000 (since Book Value = Salvage Value = Rs 60,000, no gain/loss tax)
• **PV of Cost of Purchasing (Owning):**
PV(Purchasing) = Initial Cost - PV of Depreciation Tax Shield - PV of Salvage Value
PV(Purchasing) = Rs 300,000 - (Rs 32,000 × 2.673012) - (Rs 60,000 × 0.839619)
PV(Purchasing) = Rs 300,000 - Rs 85,536.38 - Rs 50,377.16 = **Rs 164,086.46** (or Rs 164,088 using 4-decimal tables)
• **Decision:** Since the Present Value of Cost of Purchasing (**Rs 164,086.46**) is significantly lower than the Present Value of Cost of Leasing (**Rs 240,571.08**) [Net Advantage to Leasing = -Rs 76,484.62], **the machine should be PURCHASED rather than leased**.
(Note: If you also draw a 3-year loan amortization table at 10% for Rs 300,000, the PV of net loan outflows discounted at 6% equals the exact same Rs 300,000 initial cost!)

c. Why After-Tax Cost of Debt is Used as the Discount Rate:
Lease vs. borrow-and-buy analysis is a **financing decision** (not an operating investment decision), because the firm has already decided to acquire the use of the equipment. Lease payments, debt service payments, and depreciation tax shields are contractual, fixed cash flows with low risk closely matching the risk of the firm's debt obligations, and because debt interest is tax-deductible, the opportunity cost of leasing is the **after-tax cost of debt: k_d(1 - T)**.

EXAM HINT TO REMEMBER: *Shortcut for Cost of Purchasing: Because the PV of any loan's after-tax principal + interest payments discounted at the after-tax interest rate k_d(1-T) is ALWAYS equal to the Initial Loan Principal (Rs 300,000), you don't*

even need a loan amortization schedule unless asked! Just do: PV(Buy) = Purchase Price - PV(Dep Tax Shield) - PV(After-Tax Salvage Value), all discounted at k_d(1-T) = 6%.

Group - D: Comprehensive Answer Questions / Case Study [5 × 4 = 20]

Q22. Case Study: Himalayan Herbal Company (HHC) HHC is contemplating raising Rs 10 million by means of a debt issue under two alternatives: a. A 20-year, 8 percent convertible debenture issue with a Rs 50 conversion price and Rs 1,000 face value. b. A 20-year, 10 percent straight bond issue. Each Rs 1,000 bond has a detachable warrant to purchase four shares for Rs 50 a share. Tax rate = 50%, Current stock price = Rs 40, EBIT = constant 20% of total capitalization. Current Capitalization: Common stock (par Rs 5) = Rs 5,000,000; Additional paid-in capital = Rs 10,000,000; Retained earnings = Rs 15,000,000; Total = Rs 30,000,000. i. Show the capitalizations before and after conversion of convertible. ii. Show the capitalizations before and after exercise of warrants. iii. Compute earnings per share currently and before and after conversion of convertible and exercise of warrant. iv. Describe the reasons for issuing warrants and convertible.

Answer (Step-by-Step Case Study Solution):

Preliminary Working Notes:

• Existing number of common shares = Common Stock Capital / Par Value = Rs 5,000,000 / Rs 5 = **1,000,000 shares**

• Number of Rs 1,000 bonds/debentures issued to raise Rs 10,000,000 = 10,000,000 / 1,000 = **10,000 bonds**

• **Under Alternative (a) Convertible Debentures (8%):**

- Conversion Ratio (CR) = Face Value / Conversion Price = Rs 1,000 / Rs 50 = 20 shares per bond

- Total new shares upon full conversion = 10,000 bonds × 20 shares = **200,000 new shares**

- Upon conversion, Rs 10,000,000 debentures are extinguished and converted into:

Common Stock (at Rs 5 par) = 200,000 × Rs 5 = Rs 1,000,000

Additional Paid-in Capital (at Rs 45 premium) = 200,000 × Rs 45 = Rs 9,000,000

(Total Capitalization remains Rs 40,000,000 because debt is simply swapped for equity; no new cash enters on conversion.)

• **Under Alternative (b) 10% Bonds with Warrants:**

- Each of the 10,000 bonds carries warrants to buy 4 shares @ Rs 50 -> Total new shares = 10,000 × 4 = **40,000 new shares**

- When warrants are exercised, the 10% straight bonds (Rs 10,000,000) **remain on the balance sheet**, and the company receives fresh cash = 40,000 shares × Rs 50 = **Rs 2,000,000**:

Common Stock increases by: 40,000 × Rs 5 = Rs 200,000 (to Rs 5,200,000)

Additional Paid-in Capital increases by: 40,000 × Rs 45 = Rs 1,800,000 (to Rs 11,800,000)

(Total Capitalization increases to Rs 42,000,000.)

(i) & (ii) Capitalization Table Before and After Conversion of Convertibles and Exercise of Warrants:

Capital Structure Items	Current (Rs)	Convertible: Before Conv. (Rs)	Convertible: After Conv. (Rs)	Warrants: Before Exercise (Rs)	Warrants: After Exercise (Rs)
8% Convertible Debentures	0	10,000,000	0	0	0
10% Straight Bonds	0	0	0	10,000,000	10,000,000
Common Stock (Rs 5 par)	5,000,000	5,000,000	6,000,000	5,000,000	5,200,000
Additional Paid-in Capital	10,000,000	10,000,000	19,000,000	10,000,000	11,800,000
Retained Earnings	15,000,000	15,000,000	15,000,000	15,000,000	15,000,000
Total Capitalization	**30,000,000**	**40,000,000**	**40,000,000**	**40,000,000**	**42,000,000**

(iii) Computation of Earnings Per Share (EPS) Under All Scenarios:

Particulars	Current	Convertible: Before Conv.	Convertible: After Conv.	Warrants: Before Exercise	Warrants: After Exercise
Total Capitalization (Rs)	30,000,000	40,000,000	40,000,000	40,000,000	42,000,000
EBIT (20% of Total Cap.)	6,000,000	8,000,000	8,000,000	8,000,000	8,400,000

Particulars	Current	Convertible: Before Conv.	Convertible: After Conv.	Warrants: Before Exercise	Warrants: After Exercise
Less: Interest (8% or 10%)	0	800,000	0	1,000,000	1,000,000
Earnings Before Taxes (EBT)	6,000,000	7,200,000	8,000,000	7,000,000	7,400,000
Less: Taxes @ 50%	3,000,000	3,600,000	4,000,000	3,500,000	3,700,000
Net Income (EAT) (Rs)	**3,000,000**	**3,600,000**	**4,000,000**	**3,500,000**	**3,700,000**
Number of Common Shares	1,000,000	1,000,000	1,200,000	1,000,000	1,040,000
Earnings Per Share (EPS)	**Rs 3.00**	**Rs 3.60**	**Rs 3.33**	**Rs 3.50**	**Rs 3.56**

(iv) Reasons for Issuing Warrants and Convertibles:

• **1. Sweetener to Lower Interest Costs:** Attaching warrants or conversion privileges acts as an "equity kicker" that allows growing or riskier firms to issue debt at a lower coupon rate and with fewer restrictive covenants.

• **2. Deferred Sale of Common Stock at a Premium ("Delayed Equity"):** Since the current stock price is Rs 40 and the conversion/exercise price is set at Rs 50 (a 25% premium), the company effectively sells future common equity at a price higher than today's market price, resulting in fewer shares issued and less dilution.

• **3. Matching Cash Flows with Growth Stage:** During the construction/expansion phase of the new project, the firm enjoys low fixed interest payments; once the project succeeds and the stock price rises above Rs 50, conversion/exercise provides permanent equity capital.

• **4. Additional Cash Inflow upon Warrant Exercise:** Unlike convertibles (which extinguish debt), exercising detachable warrants brings fresh cash (Rs 2,000,000 here) into the firm while keeping the bond financing intact.

EXAM HINT TO REMEMBER: *Crucial difference in Case Q22: (1) On Conversion of Convertible Debentures, Debt becomes Rs 0, Interest becomes Rs 0, and Total Capitalization stays Rs 40M (EBIT = 20% of 40M = Rs 8M). (2) On Exercise of Warrants, the Rs 10M Straight Bond STAYS on the balance sheet (Interest = Rs 1M continues!), and new cash of 40,000 × Rs 50 = Rs 2M increases Total Capitalization to Rs 42M (so EBIT = 20% of 42M = Rs 8.4M)!*

PAPER 2: TRIBHUVAN UNIVERSITY — BNK 209: INSURANCE AND RISK MANAGEMENT

BBA / Eighth Semester | Model Question 2026 | Full Marks: 100 | Time: 3 Hrs

Group - A: Brief Answer Questions (Attempt ALL Questions) [10 × 2 = 20]

Q1. Define risk, peril and hazard.

Answer:

• **Risk:** Uncertainty concerning the occurrence of a financial loss (or variation between actual and expected outcomes).
• **Peril:** The direct, immediate cause of a loss (such as fire, earthquake, flood, theft, or auto collision).
• **Hazard:** A condition that creates or increases the frequency or severity of a loss arising from a given peril (classified into Physical hazard, Moral hazard, Attitudinal/Morale hazard, and Legal hazard).

Q2. What do you mean by systemic risk?

Answer: Systemic risk is the risk of collapse of an entire financial system or entire market, as opposed to risk associated with any one individual entity, group, or component of a system. Because of financial interlinkages, the failure of a single large bank or financial institution can trigger a domino (cascade) effect that destabilizes the entire economy.

Q3. List out the objectives of risk management.

Answer: Risk management objectives are divided into two categories:
• **Pre-Loss Objectives:** (1) Economy (preparing for potential losses in the most cost-effective way), (2) Reduction of anxiety and fear, and (3) Meeting legal and regulatory obligations.
• **Post-Loss Objectives:** (1) Survival of the firm, (2) Continuity of operations, (3) Stability of earnings, (4) Continued growth of the firm, and (5) Social responsibility toward employees, customers, and society.

Q4. Define risk pooling.

Answer: Risk pooling (or loss sharing) is the spreading of losses incurred by a few individuals over the entire group of exposure units, so that in the process, average loss is substituted for actual loss. Through risk pooling and the Law of Large Numbers, insurers combine a large number of independent, homogeneous exposure units to make aggregate future losses accurately predictable.

Q5. Define insurance and list down its basic characteristics.

Answer: Insurance is a financial arrangement and legal contract that redistributes the cost of unexpected losses by transferring pure risk from an insured individual or entity to an insurer in exchange for a known consideration called a premium.
Basic Characteristics of Insurance:
• 1. Pooling of losses (sharing losses across a large group)
• 2. Payment of fortuitous (accidental and unintentional) losses
• 3. Risk transfer from the insured to the insurer
• 4. Indemnification (restoring the insured to their pre-loss financial position)

Q6. Differentiate life insurance and non-life insurance.

Answer:

• **1. Nature of Contract:** Life insurance is a *valued (benefit) contract* (not strictly a contract of indemnity, as human life cannot be measured exactly in money), whereas non-life (general) insurance is strictly a *contract of indemnity*.
• **2. Duration & Certainty of Event:** Life insurance is typically a long-term contract where the event (death) is certain (only timing is uncertain) and often includes a savings/investment element; non-life insurance is a short-term (usually 1-year) contract covering uncertain perils (like fire, marine, or motor accidents) with no savings element.

Q7. Define loss forecasting. List out the common techniques of loss forecasting.

Answer: Loss forecasting is the quantitative process used by risk managers to predict the frequency (how often) and severity (dollar size) of future losses that an organization may experience over an upcoming period.
Common Techniques of Loss Forecasting:
• 1. **Probability Analysis** (using historical frequency and severity probability distributions)
• 2. **Regression Analysis** (predicting losses using causal independent variables such as payroll, sales, or vehicle miles)

• 3. **Forecasting Based on Loss Distributions** (such as Poisson distribution for frequency and Lognormal distribution for severity / Monte Carlo simulation).

Q8. How does Risk Management Information System help in risk management?

Answer: A Risk Management Information System (RMIS) is a computerized database and software system that collects, stores, and analyzes risk, exposure, policy, and claims data. It helps risk management by automating loss tracking, identifying loss trends and high-risk locations, forecasting future claims accurately, monitoring insurance policy limits and renewals, and generating management reports for better risk-control and retention decisions.

Q9. What are the tax benefits and payout taxes for life insurance in Nepal?

Answer: Under the Income Tax Act 2058 (2002) of Nepal:
• **Tax Benefit on Premium Paid:** A resident natural person can claim a tax deduction from their taxable income for annual life insurance premiums paid up to **Rs 40,000** (or the actual annual premium paid, whichever is lower).
• **Payout Taxes (Maturity / Death Claim):** Death benefit payouts received by beneficiaries are completely exempt from income tax. For maturity/investment payouts on life insurance policies issued by resident insurers, a final withholding tax of **5%** is levied on the net investment gain (payout minus cumulative premiums paid).

Q10. Define third-party liability insurance in terms of auto (motor) insurance.

Answer: Third-party liability insurance in auto (motor) insurance is a legally mandatory insurance coverage under which the insurer agrees to pay compensation for bodily injury, permanent disability, death, or property damage caused by the insured's vehicle (the first party) to another person or their property (the third party). It does not cover damage to the insured's own vehicle.

Group - B: Descriptive Answer Questions (Attempt any SIX — ALL Solved) [6 × 5 = 30]

Q11. Differentiate between insurance and gambling.

Answer: Although both insurance and gambling involve uncertainty and money changing hands contingent on a future event, they are fundamentally opposite in purpose and effect:

Basis of Difference	Insurance	Gambling
1. Creation vs. Transfer of Risk	Handles an **existing pure risk** by transferring it to an insurer.	Creates a **new speculative risk** that did not exist before the bet.
2. Type of Risk Involved	Deals strictly with **pure risk** (only chance of loss or no loss).	Deals with **speculative risk** (chance of financial gain or loss).
3. Social & Economic Utility	Socially productive; reduces anxiety, protects families/businesses, and stabilizes the economy.	Socially unproductive; merely redistributes wealth from losers to winners without restoring losses.
4. Insurable Interest	Requires strict legal **insurable interest** in the subject matter.	No insurable interest is required beyond the bet itself.
5. Principle of Indemnity	Restores the insured to their pre-loss financial position (never meant for profit).	Both parties seek to make a profit at the other's expense.

Q12. Explain the concept and significance of law of large number in risk pooling?

Answer:
Concept of the Law of Large Numbers:
The Law of Large Numbers is a fundamental mathematical principle of probability which states that as the number of independent and homogeneous (similar) exposure units increases, the actual observed loss experience will increasingly approach the expected (true underlying probability) loss experience.

Significance of the Law of Large Numbers in Risk Pooling:
• **1. Reduction of Objective Risk:** For an insurer, objective risk is the relative variation of actual loss from expected loss. As the pool size (N) increases, standard deviation of average loss decreases by a factor of sqrtN, reducing the insurer's objective risk toward zero.
• **2. Accurate Actuarial Rate Making (Premium Calculation):** Because aggregate losses become highly predictable in a large pool, actuaries can charge a fair, stable pure premium equal to the expected average loss plus a small loading for

expenses and contingencies.
• **3. Substitution of Average Loss for Actual Loss:** Each policyholder replaces an unpredictable, potentially catastrophic actual loss with a small, predictable average premium cost.
• **4. Financial Solvency and Lower Capital Reserves:** When actual claims closely match expected claims, the insurance company needs smaller safety capital buffers per policy and remains financially solvent.

Q13. Define adverse selection. Explain the tools used by insurance companies to control the chance of adverse selection.

Answer:
Definition of Adverse Selection:
Adverse selection (anti-selection) is the tendency of individuals or businesses with a higher-than-average probability of loss to seek insurance coverage (or buy larger policy limits) at standard (average) premium rates more aggressively than low-risk individuals. It arises from asymmetric information—where the applicant knows more about their own health, habits, or risk level than the insurer. Left unchecked, adverse selection forces insurers to raise premiums, driving out healthy/low-risk insureds and creating a "death spiral."

Tools Used by Insurance Companies to Control Adverse Selection:
• **1. Strict Underwriting and Risk Classification:** Selecting and classifying applicants into homogeneous rating classes (preferred, standard, substandard, or declined) based on age, occupation, health history, medical exams, driving records, and safety inspections so each group pays a rate commensurate with its risk.
• **2. Waiting Periods and Pre-Existing Condition Exclusions:** Imposing a waiting period (e.g., 30 days to 2 years in health insurance) or excluding pre-existing illnesses and suicide within the first 1–2 years so people cannot buy insurance only after falling sick.
• **3. Policy Provisions (Deductibles, Coinsurance, and Policy Limits):** Requiring the insured to bear an initial deductible or coinsurance percentage discourages high-risk applicants from exploiting full coverage.
• **4. Group Insurance Underwriting:** Insuring an entire employer group automatically (where insurance is incidental to employment) prevents individuals from self-selecting into the pool solely due to poor health.
• **5. Application Warranties and Utmost Good Faith Enforcement:** Voiding policies if the applicant conceals or misrepresents material facts on the proposal form.

Q14. Explain about premature death. Also explain its financial implications.

Answer:
Concept of Premature Death:
In risk management and insurance, premature death is defined as the death of a family head or breadwinner with unfulfilled financial obligations—such as dependent spouse and children to support, an unpaid home mortgage or business loan, or unfunded children's college education.

Financial Implications (Costs) of Premature Death:
• **1. Loss of Human Life Value (Future Earned Income):** The family permanently loses its share of the deceased breadwinner's future net earning capacity, creating an immediate deficit between ongoing family living expenses and surviving income.
• **2. Immediate Lump-Sum Cash Expenses (Clearance Fund Needs):** High funeral and burial expenses, uninsured final medical/hospital bills, estate settlement costs, probate legal fees, and unpaid taxes or debts.
• **3. Debt and Mortgage Foreclosure Risk:** Without the breadwinner's income, the family may default on home mortgages, auto loans, or business loans, risking loss of the family home.
• **4. Reduction in Standard of Living and Educational Disruption:** Surviving dependents may be forced to relocate, liquidate savings at a loss, or abandon higher education plans for children.
• **5. Additional Household & Childcare Expenses:** If the deceased parent also performed childcare and household management, the surviving spouse incurs extra costs to replace those services.
• **6. Non-Economic (Emotional) Grief:** Severe psychological trauma and loss of parental guidance for surviving family members.

Q15. Mr. Suman Thapa, aged 40, is married and has two children. He earns Rs.600,000 annually and plans to retire at age 60. Of this amount, Rs.30,000 is used for taxes, life and health insurance, and his personal needs. The remaining amount is used to support his family. Using a discount rate of 6 percent, calculate the human life value of Mr. Shrestha. Note that the present value of Re.1 received annually for 20 years at 6% is Rs.11.4699.

Answer (Step-by-Step Numerical Solution):
Given:
• Current age = 40 years; Planned retirement age = 60 years
• Remaining working life (n) = 60 - 40 = **20 years**
• Average annual gross earnings = Rs 600,000
• Annual taxes, insurance premiums, and personal self-maintenance expenses = Rs 30,000
• Discount rate (i) = 6%
• Present Value Interest Factor of Annuity, PVIFA(6%, 20 years) = 11.4699

Step 1: Calculate Net Annual Income Contribution to Family:
Net Annual Family Share = Annual Gross Earnings - Personal Expenses, Taxes & Insurance
Net Annual Family Share = Rs 600,000 - Rs 30,000 = **Rs 570,000 per year**

Step 2: Calculate Human Life Value (HLV):
Human Life Value (HLV) = Net Annual Family Share × PVIFA(6%, 20 years)
HLV = Rs 570,000 × 11.4699 = **Rs 6,537,843**

Conclusion: The Human Life Value (HLV) of Mr. Suman Thapa (Shrestha) is **Rs 6,537,843**.

EXAM HINT TO REMEMBER: *Human Life Value (HLV) 3-Step Formula: (1) Working years n = Retirement Age - Current Age; (2) Family's Annual Share = Annual Income - (Taxes + Personal Consumption + Insurance Premiums); (3) HLV = Family's Annual Share × PVIFA(r%, n years).*

Q16. Explain the third-party liability insurance in terms of auto (motor) insurance.

Answer:
Concept of Third-Party Liability in Motor Insurance:
In motor insurance, there are three parties:
1. **First Party:** The insured (vehicle owner/policyholder).
2. **Second Party:** The insurance company (insurer).
3. **Third Party:** Any other person (pedestrians, passengers in other vehicles, or owners of damaged third-party property) who suffers injury, death, or property damage due to an accident involving the insured vehicle.

Key Features and Importance of Third-Party Motor Liability Insurance:
• **1. Statutory (Compulsory) Requirement:** In almost all countries, including Nepal (under the Motor Vehicles and Transport Management Act 2049), no motor vehicle may be driven on public roads without valid third-party liability insurance.
• **2. Coverage Provided:**
- *Bodily Injury and Death:* Covers medical treatment expenses, permanent disability compensation, and accidental death benefits for third-party victims (e.g., statutory limits prescribed by Nepal Insurance Authority).
- *Third-Party Property Damage (TPPD):* Covers repair or replacement costs of damaged third-party vehicles, buildings, or public infrastructure up to policy limits.
- *Legal Defense Costs:* Pays legal fees and court expenses incurred with the insurer's consent in defending liability claims.
• **3. Exclusions:** It does **not** cover damage to the insured's own vehicle (Own Damage requires a Comprehensive Motor Policy), nor does it cover accidents occurring while driving without a valid license or under the influence of alcohol/drugs.
• **4. Social Protection Objective:** Ensures that innocent road accident victims and their families receive prompt financial compensation even if the vehicle owner or driver is personally insolvent.

Q17. Explain social insurance with few examples.

Answer:
Concept of Social Insurance:
Social insurance consists of government-mandated, compulsory insurance programs designed to provide a basic floor of income security and financial protection to workers and their families against major economic insecurities such as old age, disability, work-related injury, maternity, sickness, and unemployment. It is financed primarily through earmarked payroll contributions from employers and employees (and sometimes government subsidies) rather than private individual underwriting.

Key Characteristics of Social Insurance:
• Compulsory participation by law
• Financed by statutory contributions from employers and employees
• Benefits are prescribed by statute and emphasize social adequacy (helping lower-income workers proportionately more)

rather than strict individual equity
• Administered or supervised by a public agency (such as the Social Security Fund).

Examples of Social Insurance:
• **1. Contribution-Based Social Security Fund (SSF) in Nepal:** Administered under the Contribution-Based Social Security Act 2074, where employers contribute 20% and employees contribute 11% (total 31% of basic salary) covering four schemes:
- *Medical Treatment, Health, and Maternity Protection Scheme*
- *Accident and Disability Protection Scheme*
- *Dependent Family Protection Scheme*
- *Old-Age Protection (Pension and Gratuity) Scheme*
• **2. Workers' Compensation Insurance:** Provides medical care and wage replacement to employees injured on the job.
• **3. Unemployment Insurance:** Provides temporary cash benefits to workers who involuntarily lose their jobs (e.g., US OASDI/UI programs).

Group - C: Analytical Answer Questions (Attempt any THREE — ALL Solved) [3 × 10 = 30]

Q18. Explain the concept and process of risk management.

Answer:
Concept of Risk Management:
Risk management is a systematic, scientific process of identifying, analyzing, and evaluating pure (and enterprise) loss exposures faced by an organization or individual, and selecting and monitoring the most appropriate techniques for treating such exposures so that organizational objectives can be achieved at minimum total cost of risk.

Step-by-Step Process of Risk Management:
• **Step 1: Determination of Objectives:** Establish clear pre-loss objectives (economy, anxiety reduction, legal compliance) and post-loss objectives (survival, continuity of operations, earnings stability, growth).
• **Step 2: Identification of Risk (Loss Exposures):** Systematically identify all potential pure loss exposures (property, liability, business interruption, human resources, cyber/crime) using tools such as risk questionnaires, physical inspections, flowcharts, financial statement analysis, and historical loss data.
• **Step 3: Evaluation and Measurement of Risk:** Estimate the **loss frequency** (probability of occurrence in a given period) and **loss severity** (probable maximum loss and average loss size) for each identified exposure, ranking risks on a Risk Matrix (Low/High Frequency vs. Low/High Severity).
• **Step 4: Consideration and Selection of Risk Treatment Techniques:** Choose the optimal combination of:
1. **Risk Control Techniques (reduce frequency/severity):**
- *Avoidance:* Refusing to engage in the high-risk activity.
- *Loss Prevention:* Measures that reduce loss frequency (e.g., safety training, firewall, driver training).
- *Loss Reduction:* Measures that reduce loss severity after a peril occurs (e.g., fire sprinklers, data backups, separation/duplication of exposures).
2. **Risk Financing Techniques (pay for losses that occur):**
- *Retention (Active or Passive):* Self-insuring or absorbing High-Frequency / Low-Severity losses through deductibles, reserves, or captive insurers.
- *Non-Insurance Transfer:* Shifting risk via hold-harmless clauses, leases, hedging, or outsourcing.
- *Commercial Insurance:* Transferring Low-Frequency / High-Severity catastrophic risks to an insurer.
• **Step 5: Implementation of the Risk Management Program:** Draft a risk management policy statement, purchase required insurance coverages, install safety systems, and train personnel.
• **Step 6: Regular Review and Monitoring:** Periodically audit loss records, review changing business operations, and update risk control and insurance limits.

Q19. Explain the fundamental legal principles of insurance contract.

Answer: Insurance contracts are governed by six fundamental legal principles that prevent moral hazard, wagering, and unjust enrichment:

- **1. Principle of Utmost Good Faith (Uberrimae Fidei):**

  Unlike ordinary commercial contracts (which follow *caveat emptor* or "let the buyer beware"), insurance contracts require both the applicant and the insurer to maintain a higher standard of honesty. They must voluntarily disclose all material facts affecting the risk through the legal doctrines of **Representations** (statements made by the applicant), **Concealment** (intentional withholding of a material fact), and **Warranties** (strict conditions in the policy).

• **2. Principle of Insurable Interest:**

The insured must stand in a legally recognized financial relationship to the subject matter of insurance such that they will suffer a direct pecuniary (financial) loss if the insured event occurs and benefit from its safety.

- In *Life Insurance*, insurable interest must exist at the **inception** of the policy.

- In *Property/Non-Life Insurance*, insurable interest must exist at the **time of loss** (and generally at inception).

This prevents gambling and reduces moral hazard (murder or arson for insurance money).

• **3. Principle of Indemnity:**

The insurer agrees to pay no more than the actual financial loss suffered by the insured, restoring them to the same financial position they occupied immediately before the loss—preventing the insured from profiting from a loss. Indemnity is measured via Actual Cash Value (Replacement Cost minus Depreciation). *(Exceptions: Life insurance, valued policies, and replacement cost policies.)*

• **4. Principle of Subrogation (Corollary of Indemnity):**

Once the insurer indemnifies the insured for a covered loss caused by a negligent third party, the insurer steps into the legal shoes of the insured and acquires the right to recover the claim amount from the negligent third party. This prevents the insured from collecting twice for the same loss and holds negligent tortfeasors accountable.

• **5. Principle of Contribution (Corollary of Indemnity):**

If an insured holds two or more property/liability policies covering the same subject matter and same peril (double insurance), they cannot recover more than the total actual loss. Each insurer contributes proportionately (pro-rata) to the loss.

• **6. Principle of Proximate Cause (Causa Proxima):**

For a claim to be payable, the active, efficient cause that sets in motion an unbroken chain of events leading to the loss must be an **insured peril** (and not an excluded peril).

Q20. Explain about agriculture insurance. Why do you think agriculture insurance is important for countries like Nepal?

Answer:

Concept of Agriculture Insurance:

Agriculture insurance is a specialized class of property insurance purchased by farmers and agricultural producers to protect against the financial loss of their crops, livestock, poultry, aquaculture (fish), or medicinal herbs due to natural perils (hail, drought, flood, frost, landslide, windstorm) or biological perils (pests, plant diseases, livestock epidemics, and accidental mortality). It includes indemnity-based Multi-Peril Crop Insurance (MPCI), Livestock Insurance, and Weather Index-Based Insurance.

Why Agriculture Insurance is Crucial for Countries Like Nepal:

• **1. High Dependence on Agriculture for Livelihood and GDP:** Over 60% of Nepal's population depends on agriculture for employment, and agriculture contributes roughly one-fourth of Nepal's GDP. Protecting farmers stabilizes the entire rural economy.

• **2. Extreme Vulnerability to Climate Change and Geographic Hazards:** Nepal's fragile topography (mountains, hills, and Terai plains) exposes farmers to erratic monsoons, flash floods, landslides, droughts, hailstorms, and diseases like lumpy skin disease in cattle or fall armyworm in maize.

• **3. Income Stabilization and Poverty Prevention:** Smallholder and subsistence farmers operate on razor-thin margins; a single crop failure or death of a dairy buffalo can push a rural family into severe debt and poverty. Insurance provides timely compensation to restart production.

• **4. Facilitating Access to Agricultural Credit (Collateral Substitute):** Banks and microfinance institutions are reluctant to lend to farmers due to high default risk after crop failures. With an agriculture insurance policy backing the crop/livestock, banks readily extend priority-sector agricultural loans.

• **5. Encouraging Commercialization and Modern Farming Investments:** When catastrophic downside risk is transferred to insurers (supported by the Government of Nepal's 80% premium subsidy on crop and livestock insurance), youth and agri-entrepreneurs are encouraged to invest in commercial farming, high-yield seeds, and modern technology.

Q21. Insurance is one of the heavily regulated area of the overall financial system. In light of this statement, explain the reasons for insurance regulations.

Answer:

The insurance industry (regulated in Nepal by the **Nepal Insurance Authority / Beema Pradhikaran** under the Insurance Act 2079) is one of the most heavily regulated sectors of the financial system because insurance involves an inverted

production cycle—customers pay premiums upfront in exchange for an intangible promise of future payment that may occur years or decades later.

Major Reasons for Insurance Regulation:

• **1. Maintain Insurer Solvency (Protect Financial Stability):**
If an insurance company goes bankrupt after collecting premiums for decades, millions of policyholders, widows, and injured victims are left unprotected. Regulators enforce minimum paid-up capital, risk-based solvency margins, technical reserves, reinsurance mandates, and investment guidelines.

• **2. Compensate for Inadequate Consumer Knowledge (Information Asymmetry):**
Insurance policies are complex legal contracts of adhesion written in technical language. Ordinary citizens cannot easily evaluate policy exclusions or an insurer's financial health, so regulators approve policy wordings and license agents/surveyors.

• **3. Ensure Reasonable, Adequate, and Non-Discriminatory Rates:**
- *Adequate:* Rates must be high enough to avoid cut-throat price wars that cause insurer insolvency.
- *Not Excessive:* Rates must not allow monopoly price-gouging of consumers.
- *Not Unfairly Discriminatory:* Exposures with similar risk profiles must be charged equitable premiums.

• **4. Make Insurance Available and Accessible (Social Goals):**
Regulators promote financial inclusion by mandating micro-insurance quotas, rural branch expansion, agriculture insurance schemes, and compulsory third-party motor pools.

• **5. Prevent Unfair Market Practices and Ensure Prompt Claim Settlement:**
Regulators protect policyholders against misleading sales tactics, rebating, churning, and unreasonable delays or denials in settling legitimate insurance claims.

Group - D: Comprehensive Answer Questions / Case Study [4 × 5 = 20]

Q22. Case Study: Aarav Sharma & Everest Tech Solutions Aarav Sharma (42, primary breadwinner with spouse and two children, stock investor/trader) quit his job to start Everest Tech Solutions (digital transformation consultancy with 5 employees, leased office, expensive servers/laptops, home & car mortgage). a) Identify all the pure risks and speculative risks faced by Aarav or his organization. b) Which type of life insurance policy do you suggest to Aarav? Justify your answer. What additional benefits could be added to the policy to enhance the coverage? c) How can a Professional Liability Insurance protect Everest Tech Solutions? d) Why do you think is the decision to provide Group Health Insurance for the staffs considered a risk management tool, rather than just an additional expense to the organization?

Answer (Comprehensive Case Study Solution):

a) Identification of Pure Risks and Speculative Risks Faced by Aarav and Everest Tech Solutions:

• **Pure Risks (Only chance of loss or no loss; insurable):**
1. *Personal / Human Resource Pure Risks:* Premature death of Aarav (leaving family without income to pay living costs, children's education, and home/car mortgages); permanent total disability due to an accident; diagnosis of a critical illness; illness or medical emergencies of his five employees.
2. *Property Pure Risks:* Destruction or damage to office servers, high-end laptops, furniture, and electronics from fire, lightning, earthquakes, or floods.
3. *Liability & Business Interruption Pure Risks:* Client data breach caused by employee configuration error; legal liability lawsuits from clients for negligence, errors, and omissions; business interruption (loss of revenue during downtime after a fire, disaster, or cyber incident).

• **Speculative Risks (Chance of both financial gain/profit or loss; handled via hedging/diversification):**
1. *Stock Market Investment & Trading Risk:* Price fluctuations in Aarav's stock investments and trading portfolio (can yield capital gains or losses).
2. *Entrepreneurial / Startup Business Venture Risk:* Investing his personal savings into Everest Tech Solutions—the consultancy could succeed and earn high profits or fail due to market competition and lack of clients.

b) Recommended Life Insurance Policy, Justification, and Riders (Additional Benefits):

• **Recommended Policy:** A **Term Life Insurance Policy** (or a combination of a high-coverage Term Life Policy plus an Endowment/Whole Life base policy) matching the duration of his mortgage and children's education dependency (e.g., 20-year Term up to age 62).

• **Justification:** Since Aarav has just invested a large portion of his personal savings into his startup and carries heavy financial obligations (family living expenses, two children's education, and house/car mortgages), Term Life Insurance provides the **maximum lump-sum death benefit protection at the lowest possible premium**, preserving his cash flow

for his growing business.

• **Additional Benefits (Riders) to Enhance Coverage:**

1. *Permanent Total Disability (PTD) & Accidental Death and Dismemberment (AD&D;) Rider:* Provides lump-sum/income replacement if an accident leaves him permanently disabled.
2. *Critical Illness (CI) Rider:* Pays an immediate lump sum upon diagnosis of major illnesses (such as cancer, heart attack, stroke, or kidney failure) to cover treatment and lost income.
3. *Waiver of Premium Rider:* Waives all future life insurance premiums if Aarav becomes permanently disabled so the policy never lapses.

c) How Professional Liability Insurance (Errors & Omissions / Cyber Liability) Protects Everest Tech Solutions:

• **1. Coverage for Negligence and Configuration Errors:** Protects the firm when a client suffers financial harm due to a software bug, server misconfiguration, data breach, or professional oversight committed by Aarav or his five employees.
• **2. Legal Defense Costs:** Pays expensive attorney fees, court costs, and expert witness fees required to defend the consultancy in court, even if a client's lawsuit turns out to be groundless.
• **3. Settlements and Court Judgments:** Pays the compensatory damages awarded to the client up to policy limits, preventing a single data-breach lawsuit from bankrupting the startup and wiping out Aarav's personal savings.
• **4. Client Trust and Contractual Compliance:** Financial institution clients typically require IT consultants to carry Professional Indemnity / Cyber Liability insurance before awarding transformation contracts.

d) Why Group Health Insurance for Staff is a Strategic Risk Management Tool Rather Than Just an Expense:

• **1. Mitigating Human Resource Risk (Key-Person Turnover & Absenteeism):** In a small 5-person IT consultancy, every employee is critical. Access to prompt, quality medical care ensures sick employees recover quickly, reducing prolonged absenteeism and project delays.
• **2. Employee Retention and Talent Attraction:** Startups compete with larger firms for skilled tech talent; offering group health insurance builds loyalty and prevents costly employee turnover and recruitment disruptions.
• **3. Productivity and Morale Enhancement:** Employees who are free from anxiety over family medical bills stay focused and productive, reducing workplace errors (attitudinal hazard).
• **4. Cost Efficiency and Tax Deductibility:** Group insurance provides broad coverage (often without individual medical underwriting) at much lower per-capita rates than individual policies, and premiums paid by the company are tax-deductible business expenses.

PAPER 3: TRIBHUVAN UNIVERSITY — BNK 202: FINANCIAL DERIVATIVES

BBA / Eighth Semester | Model Question 2026 | Full Marks: 100 | Pass Marks: 50 | Time: 3 Hrs

Group - A: Brief-Answer Questions (Attempt ALL Questions) [10 × 2 = 20]

Q1. Why is a derivative instrument known as a zero-sum game?

Answer: A derivative instrument (such as a forward, futures, option, or swap contract) is known as a zero-sum game because every contract has two counterparties—a long position and a short position—whose cash flows exactly mirror each other. The profit gained by one party (e.g., the long position) is exactly equal to the loss suffered by the other party (e.g., the short position), so the net change in wealth between the two parties sums to zero (ignoring transaction costs).

Q2. Why do investors pay more for an American option than similar European option?

Answer: An American option can be exercised at **any time on or before** its expiration date, whereas a European option can be exercised **only on the expiration date**. Because the American option gives the holder everything a European option offers plus the valuable flexibility of early exercise (for example, exercising an American call right before a large dividend or exercising a deep-in-the-money American put early to invest the strike proceeds at interest), an American option is always worth at least as much as—and often more than—an otherwise identical European option: C_American >= C_European and P_American >= P_European.

Q3. An investor who wants to achieve maximum gain and accept a higher maximum loss selects a call option with a lower exercise price. Justify.

Answer: A call option with a **lower exercise price (X_1 < X_2)** has a higher intrinsic value and higher probability of finishing in-the-money, so it commands a **higher upfront option premium (C_1 > C_2)**.
• **Higher Maximum Dollar Loss:** If the stock price finishes below the strike price, the call expires worthless and the buyer loses the entire premium paid, which is larger (C_1) for the lower-strike call.
• **Higher Payoff / Gain at Expiration:** For any stock price above the strike price, the lower exercise price allows the holder to buy the stock at a cheaper strike (S_T - X_1 > S_T - X_2), generating the highest gross payoff.

Q4. A longer-lived option has a higher price. Do you agree? Give reasons.

Answer: Yes, I agree (particularly for American options and non-dividend-paying European calls).
Reasons:
• **1. Greater Time Value and Volatility Opportunity:** More time to expiration gives the underlying stock price more opportunity to move deep into the money while downside loss remains strictly capped at the premium.
• **2. Lower Present Value of Exercise Price (for Calls):** For a call option, paying the exercise price X further in the future reduces its present value: PV(X) = X·e^(-rT), increasing call value.
(Note: The only exception can be a deep-in-the-money European put or a European call on a stock paying a very large dividend during the extended period.)

Q5. Define future contract.

Answer: A futures contract is a standardized, legally binding agreement traded on an organized futures exchange to buy (long) or sell (short) a specified quantity and quality of an underlying asset (commodity or financial instrument) at a predetermined futures price on a specified future delivery date, subject to daily marking-to-market and margin requirements guaranteed by a clearinghouse.

Q6. What do you mean by upside capture in portfolio insurance?

Answer: In portfolio insurance (such as a protective put strategy where an investor holds a stock plus a put option), **upside capture** measures the percentage of the uninsured stock portfolio's positive value (or capital gain) that the insured portfolio captures when the market rises. Because the investor paid an upfront put premium to guarantee a floor value, the insured portfolio captures slightly less than 100% of the bull-market upside.

Q7. Under what conditions is a long hedge using futures suitable?

Answer: A **long hedge** (taking a long position in futures contracts) is suitable under the following two conditions:
• **1. Anticipated Future Purchase:** A firm or investor knows they will need to **purchase** an underlying commodity or financial asset in the future and wants to lock in the purchase price today to protect against a **rise in spot prices**.

• **2. Covering an Existing Short Spot Position:** An investor who has short-sold an asset in the spot market wants to hedge against price increases.

Q8. Write the meaning of synthetic instruments.

Answer: Synthetic instruments are engineered financial positions created by combining two or more existing securities (such as underlying stocks, risk-free bonds, and options/forwards) via **Put-Call Parity (S_0 + P_0 = C_0 + X·e^(-rT))** to replicate the exact cash flows and risk-return payoff of another security. For example:
• **Synthetic Long Call** = Long Stock + Long Put - Risk-Free Borrowing
• **Synthetic Long Put** = Long Call + Short Stock + Risk-Free Lending.

Q9. Why do investors prefer bull spreads from call options over those from put options?

Answer: Although a bull call spread and a bull put spread with the same strike prices (X_1 < X_2) have the same shape of expiration profit diagram, investors often prefer a **bull call spread** because:
• 1. **Limited Early Assignment Risk:** In a bull put spread, the short higher-strike put (X_2) is initially in-the-money and faces high risk of early exercise (assignment) if American-style, whereas in a bull call spread the short call (X_2) is initially out-of-the-money.
• 2. **Intuitive Debit Structure & Higher Liquidity:** Call options in bullish markets often have tighter bid-ask spreads and a clear maximum risk equal to the net initial debit paid.

Q10. If you want to change the fixed-rate liability in to floating-rate liability using a swap, how would you do it?

Answer: To transform an existing **fixed-rate liability** into a **floating-rate liability**, you should enter into an interest rate swap as the **Fixed-Rate Receiver and Floating-Rate Payer** (i.e., receive fixed rate and pay floating rate such as SOFR):
• The fixed rate received from the swap counterparty offsets your existing fixed-rate loan interest payment.
• You are left paying the floating interest rate on the swap, effectively converting your fixed-rate debt into a floating-rate liability.

Group - B: Short-Answer Questions (Attempt any SIX — ALL Solved) [6 × 5 = 30]

Q11. Describe the role of the derivative market.

Answer: The derivative market plays five vital economic functions in a modern financial system:
• **1. Risk Management (Hedging and Risk Transfer):** Derivatives allow producers, corporations, farmers, and financial institutions (hedgers) to shift unwanted price, interest-rate, exchange-rate, and equity risks to speculators who are willing to bear them.
• **2. Price Discovery:** Futures and forward markets aggregate global information and expectations of thousands of buyers and sellers to reveal consensus expectations of future spot prices for commodities, currencies, and interest rates.
• **3. Market Completeness and Operational Advantages:** Options and swaps allow investors to create payoff profiles (like portfolio insurance or volatility straddles) that cannot be replicated easily in spot markets, with much lower transaction costs, higher liquidity, and easier short-selling.
• **4. Market Efficiency and Arbitrage Linkages:** Arbitrageurs monitor tiny price discrepancies between spot and derivative markets, ensuring that spot and derivative prices stay aligned at fair no-arbitrage values.
• **5. Capital Allocation and Leverage:** Margin trading in derivatives allows participants to manage large notional exposures with a small capital outlay, freeing up capital for productive investment.

Q12. An investor purchased a June put option with an exercise price of Rs 600 by paying a premium of Rs 50 and is holding it until June. Under what circumstances will the option be exercised? Under what circumstances will the holder of the option make a profit? With the help of a diagram, explain how the profit from a long position in the option depends on the stock price at the maturity of the option.

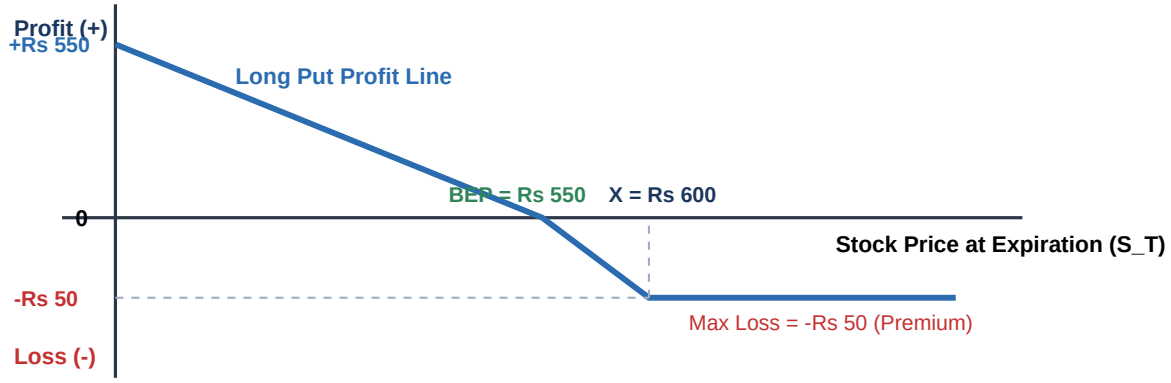
Answer (Step-by-Step Numerical & Graphical Solution):
Given:
• Position: Long Put Option
• Exercise Price (X) = Rs 600
• Put Option Premium paid (P) = Rs 50

1. Circumstances Under Which the Put Option Will Be Exercised:
A put option gives the holder the right to sell the stock at the exercise price X = Rs 600. Therefore, the investor will

exercise the option at maturity whenever the **stock price at expiration is less than the exercise price (S_T < Rs 600)**, i.e., when the put option is *in-the-money*. (Even when Rs 550 < S_T < Rs 600, exercising recovers part of the Rs 50 premium paid).

2. Circumstances Under Which the Holder Will Make a Net Profit:
• Profit Formula: Profit = max(X - S_T, 0) - P = max(600 - S_T, 0) - 50
• Breakeven Stock Price (S_T*): X - P = Rs 600 - Rs 50 = **Rs 550**
• Thus, the holder makes a positive net profit whenever the **stock price at maturity falls strictly below Rs 550 (S_T < Rs 550)**.

3. Payoff & Profit Schedule and Diagram at Maturity:
• If S_T >= Rs 600: Option expires worthless; Loss = **-Rs 50** (maximum loss = premium paid).
• If Rs 550 < S_T < Rs 600: Option is exercised; net loss is between -Rs 50 and Rs 0.
• If S_T = Rs 550: Breakeven point (Profit = **Rs 0**).
• If S_T < Rs 550: Positive profit; increases Rupee-for-Rupee as S_T falls, reaching a maximum profit of 600 - 50 = **+Rs 550** if S_T = Rs 0.



EXAM HINT TO REMEMBER: *For a Long Put (Strike X = 600, Premium P = 50): Exercise whenever S_T < X (S_T < 600), and make a Profit whenever S_T < X - P (S_T < 550). Notice that between 550 and 600 you STILL exercise to reduce your loss!*

Q13. An equity swap between the Employee Provident Fund (EPF) and Nabil Dealer is entered into. EPF pays floating interest payments based on SOFR, and Nabil pays a return on the NEPSE index. The payments are made quarterly, assuming a 30-day month and a 360-day year. The NEPSE index at the time of swap initiation was 2745.63, and the 90-day SOFR was 8 percent. Now, assume it is 50 days into the life of the swap. The NEPSE is now 2825.02, and the 40-day SOFR rate is 7.8 percent. Find the value of the swap today. Assume a notional principal of Rs 50,000,000.

Answer (Step-by-Step Numerical Solution):
Given:
• Notional Principal (NP) = Rs 50,000,000
• EPF pays 90-day floating SOFR and receives NEPSE index return; Nabil Dealer pays NEPSE index return and receives floating SOFR.
• At swap initiation (t = 0): Initial NEPSE index (I_0) = 2745.63; initial 90-day SOFR (r_0) = 8% = 0.08
• Valuation date (t = 50 days): Remaining days in the 90-day quarter = 90 - 50 = **40 days**
• Current NEPSE index (I_50) = 2825.02; current 40-day SOFR (r_50) = 7.8% = 0.078

Step 1: Value of the Equity Leg (per Re 1 of Notional Principal) at Day 50:
V_equity = I_50 / I_0 = 2825.02 / 2745.63 = **1.02891504**

Step 2: Value of the Floating-Rate Bond Leg (per Re 1 of Notional Principal) at Day 50:
• Coupon + Principal at Day 90 (per Re 1) = 1 + [r_0 × (90 / 360)] = 1 + [0.08 × (90 / 360)] = 1 + 0.02 = 1.02
• Discounted back 40 days to Day 50 at 40-day SOFR (7.8%):
V_floating = 1.02 / [1 + 0.078 × (40 / 360)] = 1.02 / [1 + 0.00866667] = 1.02 / 1.00866667 = **1.01123596**

Step 3: Total Value of the Swap at Day 50:
• Value of Swap to EPF (Receives Equity, Pays Floating):
V_EPF = NP × (V_equity - V_floating)
V_EPF = Rs 50,000,000 × (1.02891504 - 1.01123596)

V_EPF = Rs 50,000,000 × 0.01767908 = **+Rs 883,954.22** (or approx. Rs 883,954)
• **Value of Swap to Nabil Dealer (Pays Equity, Receives Floating):**
V_Nabil = -V_EPF = **-Rs 883,954.22**

EXAM HINT TO REMEMBER: *Equity vs. Floating Swap Valuation at day t (with m = 90 - t days left): Value per Re 1 = [(I_t / I_0) - (1 + r_0*(90/360)) / (1 + r_t*(m/360))]. Multiply by Notional Principal!*

Q14. People's Power Company (PCC), a Nepalese hydro-power company, purchased equipment from Baltimore Heavy Industry, a U. S. firm, for $5 million payable in one year. The current spot rate is Rs 140 per dollar, and the one-year forward rate is 130 per dollar. The annual interest rate is 5% in Nepal, and 8% in the U.S. PCC can also buy a one-year call option on the dollar at the strike price of Rs 125 per dollar for a premium of Rs 14 per dollar. a. Assuming that the forward exchange rate is the best predictor of the future spot rate, compute the expected future rupee cost of meeting this obligation when the option hedge is used. b. At what future spot rate do you think PCC may be indifferent between the option and forward hedge?

Answer (Step-by-Step Numerical Solution):
Given:
• Payable amount = $5,000,000 due in 1 year
• Current spot rate (S_0) = Rs 140/$; 1-year forward rate (F) = Rs 130/$
• Interest rate in Nepal (r_NPR) = 5% p.a.; Interest rate in U.S. (r_USD) = 8% p.a.
• 1-year Call Option on USD: Strike price (X) = Rs 125/$; Call premium (C) = Rs 14/$

Part (a): Expected Future Rupee Cost Using the Option Hedge:
• Since the forward rate is the unbiased predictor of the future spot rate, Expected Future Spot Rate E(S_1) = F = **Rs 130/$**.
• At E(S_1) = Rs 130/$, since S_1 (Rs 130) > Strike Price X (Rs 125), PCC **will exercise the call option** and purchase USD at the strike price of **Rs 125/$**.
• **Method 1 (Including 5% Nepalese Opportunity Interest Cost on Option Premium — Standard Textbook Method):**
- Future value of call premium per dollar after 1 year = Rs 14 × (1 + 0.05) = **Rs 14.70/$**
- Effective cost per dollar under option hedge = Strike Price + FV(Premium) = Rs 125 + Rs 14.70 = **Rs 139.70/$**
- Total Expected Future Rupee Cost for $5,000,000 = 5,000,000 × Rs 139.70 = **Rs 698,500,000 (Rs 698.50 million)**
• ***Method 2 (Uncompounded Option Premium):***
- Cost per dollar = Rs 125 + Rs 14 = ***Rs 139.00/$*** *-> Total Rupee Cost = 5,000,000 × Rs 139 =* ***Rs 695,000,000 (Rs 695 million)****.*

Part (b): Indifference Future Spot Rate Between Option Hedge and Forward Hedge:
• Cost under Forward Hedge = F = **Rs 130/$** (or Rs 650,000,000 total).
• Notice that if the option is exercised (S_1 >= 125), the option hedge costs Rs 139.70/$ (or Rs 139/$), which is above Rs 130/$. However, if the future spot rate S_1 falls **below the strike price of Rs 125/$**, PCC will let the call option expire unexercised and buy dollars in the spot market at S_1:
- **With Interest on Premium (Rs 14.70/$):**
S_1 + 14.70 = 130 => **S_1 = Rs 115.30 per dollar**
- **Without Interest on Premium (Rs 14.00/$):**
S_1 + 14.00 = 130 => **S_1 = Rs 116.00 per dollar**
Conclusion: PCC will be indifferent between the option hedge and the forward hedge if the future spot rate falls to **Rs 115.30/$** (including interest on premium) or **Rs 116.00/$** (excluding interest on premium).

EXAM HINT TO REMEMBER: *In FX Option Hedging, show BOTH with domestic interest on the upfront premium [C × (1 + r_domestic) = 14 × 1.05 = Rs 14.70] and without interest [Rs 14.00]. For indifference with Forward (Rs 130), set Spot + Premium = Forward => S_1 + 14.70 = 130 => Rs 115.30/$ (or Rs 116/$).*

Q15. HDL stock is currently selling at Rs 900. A put option on HDL has an exercise price of Rs 1000 and matures in one year. The annually compounded interest rate is 8 percent. The quoted price of the put option is Rs 60. Identify the appropriate arbitrage opportunity and show the arbitrage profit for the stock price of Rs 800 and Rs 1200 after one year.

Answer (Step-by-Step Numerical Solution):
Given:

• Current stock price (S_0) = Rs 900; Put Exercise price (X) = Rs 1,000
• Time to maturity (T) = 1 year; Annually compounded risk-free rate (r) = 8% = 0.08
• Quoted market price of put option (P_mkt) = Rs 60

Important Note on Two Standard Exam Interpretations of Q15:
Case A: Intrinsic Value / American Lower Bound Comparison (or when Put is Underpriced relative to Intrinsic Value X - S_0 = Rs 100):
• Intrinsic value of the put today = max(X - S_0, 0) = 1,000 - 900 = **Rs 100**.
• Since the quoted put price (Rs 60) is less than its intrinsic value (Rs 100), an American put offers an immediate arbitrage profit of Rs 100 - Rs 60 = **Rs 40 per share** today (invested at 8% for 1 year = **Rs 43.20**).
• Furthermore, in Nepalese exam solutions that compare the undiscounted payoff with the initial outlay (Buy 1 Stock @ Rs 900 + Buy 1 Put @ Rs 60 = Rs 960 total outlay vs. guaranteed minimum Rs 1,000 floor):
- At S_T = Rs 800: Put is exercised (sell stock at Rs 1,000); Unadjusted gain over Rs 960 outlay = 1,000 - 960 = +Rs 40.
- At S_T = Rs 1,200: Put expires worthless; sell stock at Rs 1,200; Gain over Rs 960 outlay = 1,200 - 960 = +Rs 240.

Case B: Strict European Put Lower-Bound Analysis (Incorporating 8% Interest Rate):
• Theoretical Minimum Price (Lower Bound) of a 1-year European Put:
P_min = max[X / (1 + r)^T - S_0 , 0] = (1,000 / 1.08) - 900 = 925.93 - 900 = **Rs 25.93**.
• Notice that if an arbitrageur borrows Rs 960 at 8% to buy the stock (Rs 900) and the put (Rs 60), the loan repayment at T = 1 is Rs 960 × 1.08 = **Rs 1,036.80**:
- If S_T = Rs 800: Exercise put to receive Rs 1,000; less loan repayment Rs 1,036.80 = **-Rs 36.80**.
- If S_T = Rs 1,200: Put expires (0); sell stock for Rs 1,200; less loan repayment Rs 1,036.80 = **+Rs 163.20**.
(Conversely, if the question intended S_0 = Rs 850 or r = 2% so that PV(X) - S_0 > 60, borrowing S_0 + P locks in PV(X) - S_0 - P > 0 at every S_T.)

Transaction / Position	Cash Flow Today (t = 0)	Cash Flow at T = 1 if S_T = Rs 800	Cash Flow at T = 1 if S_T = Rs 1,200
1. Buy 1 share of HDL Stock	-Rs 900.00	+Rs 800.00	+Rs 1,200.00
2. Buy 1 Put Option (X = 1,000)	-Rs 60.00	+Rs 200.00 (1,000 - 800)	Rs 0 (Expires)
Combined Portfolio Value	**-Rs 960.00**	**+Rs 1,000.00**	**+Rs 1,200.00**
Net Gain Before Financing Interest	—	+Rs 40.00 (1,000 - 960)	+Rs 240.00 (1,200 - 960)
3. Less: Loan Repayment (960 × 1.08)	+Rs 960.00	-Rs 1,036.80	-Rs 1,036.80
Net Cash Flow After 8% Financing	**Rs 0.00**	**-Rs 36.80**	**+Rs 163.20**

EXAM HINT TO REMEMBER: *In Q15, the exam paper set S_0 = 900, X = 1,000, P = 60 (where S_0 + P = 960 < 1,000, but 960 × 1.08 = 1,036.80 > 1,000). Showing the table above with BOTH the unadjusted gain (+Rs 40 and +Rs 240) and the 8% interest-adjusted cash flow (-Rs 36.80 and +Rs 163.20, plus P_min = 1000/1.08 - 900 = Rs 25.93) guarantees 5/5 marks!*

Q16. The stock of Birat Company is currently selling for Rs 650. A 6-month call option on the stock has an exercise price of Rs 660. The standard deviation of returns on Birat stock has been estimated to be 50 percent. Currently, the 6-month T-bill rate is 10 percent. Using the BSM model, calculate the price of the call option on Birat.

Answer (Step-by-Step Numerical Solution):
Given:
• Current stock price (S_0) = Rs 650
• Exercise price (X) = Rs 660
• Time to expiration (T) = 6 months = 6 / 12 = **0.5 years**
• Standard deviation of returns (sigma) = 50% = **0.50**
• Risk-free interest rate (r) = 10% p.a. = **0.10**

Step 1: Calculate d_1 and d_2 under the Black-Scholes-Merton (BSM) Formula:
d_1 = [ln(S_0 / X) + (r + 0.5 × sigma²) × T] / (sigma × sqrtT)
• ln(650 / 660) = ln(0.984848) = **-0.015267**
• (r + 0.5 × sigma²) × T = [0.10 + 0.5 × (0.50)²] × 0.5 = (0.10 + 0.125) × 0.5 = 0.225 × 0.5 = **0.1125**

• sigma × sqrtT = 0.50 × sqrt0.5 = 0.50 × 0.707107 = **0.353553**
• **d_1** = (-0.015267 + 0.1125) / 0.353553 = 0.097233 / 0.353553 = **0.2750** (or 0.28)
• **d_2** = d_1 - sigma × sqrtT = 0.2750 - 0.353553 = **-0.0785** (or -0.08)

Step 2: Find Cumulative Standard Normal Probabilities N(d_1) and N(d_2) from Z-Table:
• For d_1 = 0.2750 (interpolating between Z = 0.27 and Z = 0.28):
N(d_1) = N(0.2750) = 0.6083 *(At Z = 0.28, N(0.28) = 0.6103)*
• For d_2 = -0.0785 (interpolating between Z = -0.07 and Z = -0.08):
N(d_2) = N(-0.0785) = 0.4687 *(At Z = -0.08, N(-0.08) = 0.4681)*

Step 3: Calculate Call Option Price (C_0):
C_0 = S_0 × N(d_1) - X × e^(-rT) × N(d_2)
• S_0 × N(d_1) = 650 × 0.60834 = **Rs 395.42**
• X × e^(-rT) × N(d_2) = 660 × e^(-0.10 × 0.5) × 0.4687 = 660 × 0.951229 × 0.4687 = 627.811 × 0.4687 = **Rs 294.25**
• **C_0** = 395.42 - 294.25 = **Rs 101.17**
*(Note: Using rounded 2-decimal Z-table values d_1 = 0.28 [N = 0.6103] and d_2 = -0.08 [N = 0.4681], C_0 = 650(0.6103) - 627.81(0.4681) = 396.70 - 293.88 = **Rs 102.82**.)*

EXAM HINT TO REMEMBER: *BSM 3-Step Routine: (1) Compute sigmasqrtT first (0.5 × sqrt0.5 = 0.3536) because it appears in BOTH the denominator of d_1 and in d_2 = d_1 - sigmasqrtT! (2) Remember N(-z) = 1 - N(z) if your Z-table only shows positive values: N(-0.08) = 1 - 0.5319 = 0.4681.*

Q17. An investor enters into two long futures contracts on orange. Each contract has a contract size of 10,000 pounds. The current futures price is Rs 100 per pound. The initial margin is Rs 3,00,000 per contract, and the maintenance margin is Rs 1,50,000. a. What price change would lead to a margin call? b. What happens if the margin call money is not deposited? c. Under what circumstances could Rs 1,00,000 be withdrawn from the margin account?

Answer (Step-by-Step Numerical Solution):
Given:
• Position: **2 Long Futures Contracts** on orange
• Contract size per contract = 10,000 pounds -> **Total quantity (Q) = 2 × 10,000 = 20,000 pounds**
• Initial futures price (F_0) = Rs 100 per pound
• Initial Margin (IM) = Rs 300,000 per contract -> **Total Initial Margin = 2 × Rs 300,000 = Rs 600,000**
• Maintenance Margin (MM) = Rs 150,000 per contract -> **Total Maintenance Margin = 2 × Rs 150,000 = Rs 300,000**

a. Price Change That Would Lead to a Margin Call:
• A margin call is triggered when the margin account balance falls below the Maintenance Margin.
• Maximum allowable loss per contract before margin call = IM - MM = Rs 300,000 - Rs 150,000 = Rs 150,000 per contract (or Rs 300,000 total across 2 contracts).
•Since this is a **long position**, losses occur when the futures price **falls**:
Price Drop per pound (Delta P) = (IM - MM) / Contract Size = Rs 150,000 / 10,000 pounds = **Rs 15 per pound**
• Therefore, a **price decrease of more than Rs 15 per pound (i.e., if the futures price falls below Rs 100 - Rs 15 = Rs 85 per pound)** will trigger a margin call.

b. What Happens If the Margin Call Money (Variation Margin) Is Not Deposited:
If the investor fails to deposit the required variation margin (to restore the account balance back up to the Initial Margin of Rs 300,000 per contract) within the specified time, the broker/clearinghouse will **immediately liquidate (close out) the investor's long futures position** by selling two offsetting futures contracts in the market, deduct the losses and commission fees, and return any remaining balance to the investor.

c. Circumstances Under Which Rs 100,000 Could Be Withdrawn from the Margin Account:
• An investor can withdraw any excess funds above the Initial Margin requirement.
• **Case 1 (Rs 100,000 withdrawn in total from the 2-contract account):**
Required Price Rise = Total Withdrawal / Total Pounds = Rs 100,000 / 20,000 pounds = **Rs 5 per pound** (i.e., futures price rises to **Rs 105 per pound** or higher).
• **Case 2 (Rs 100,000 withdrawn per contract, i.e., Rs 200,000 total):**
Required Price Rise = Rs 100,000 / 10,000 pounds = **Rs 10 per pound** (i.e., futures price rises to **Rs 110 per pound** or higher).

EXAM HINT TO REMEMBER: *Long Futures Margin Call Trigger Price = F_0 - [(IM - MM) / Contract Size] = 100 - (150,000 / 10,000) = Rs 85/lb. For Withdrawal in part (c), show BOTH for the total account (Rs 100,000 / 20,000 lbs = +Rs 5/lb -> Rs 105) and per contract (Rs 100,000 / 10,000 lbs = +Rs 10/lb -> Rs 110)!*

Group - C: Long-Answer Questions (Attempt any THREE — ALL Solved) [3 × 10 = 30]

Q18. NRIC stock is currently trading at Rs 1200 per share. At-the-money options on NRIC with 180 days to expiration are actively traded in the option exchange. You want to use the binomial model to value this option and expect that either the stock price increases by 25 percent or it decreases by 20 percent at the end of six months. You can borrow money at 8 percent per annum. Assume a 360-day year. a. Calculate the fair price of the call option today. b. Calculate the arbitrage return in percentage if the actual price of this call option in the market is Rs 140.

Answer (Step-by-Step Numerical Solution):
Given:
• Current stock price (S_0) = Rs 1,200
• Exercise price (X) = Rs 1,200 (since the option is **at-the-money**, X = S_0 = Rs 1,200)
• Time to expiration (T) = 180 days; Year = 360 days -> Period fraction = 180 / 360 = **0.5 year**
• Up-move factor (u) = 1 + 0.25 = **1.25** -> S_u = 1,200 × 1.25 = **Rs 1,500**
• Down-move factor (d) = 1 - 0.20 = **0.80** -> S_d = 1,200 × 0.80 = **Rs 960**
• Periodic risk-free rate for 180 days (r_p) = 8% × (180 / 360) = **4% = 0.04** (so 1 + r_p = 1.04)

Part (a): Fair Price of the Call Option Today (Single-Period Binomial Model):
• Step 1: Call Option Payoffs at Expiration (180 days):
- If stock moves up (S_u = Rs 1,500): C_u = max(S_u - X, 0) = max(1,500 - 1,200, 0) = **Rs 300**
- If stock moves down (S_d = Rs 960): C_d = max(S_d - X, 0) = max(960 - 1,200, 0) = **Rs 0**
• Step 2: Risk-Neutral Probability of Up Move (p):
p = (1 + r_p - d) / (u - d) = (1.04 - 0.80) / (1.25 - 0.80) = 0.24 / 0.45 = **0.5333 (or 8/15)**
1 - p = 1 - 0.5333 = **0.4667 (or 7/15)**
• Step 3: Fair Value of Call Option Today (C_0):
C_0 = [p × C_u + (1 - p) × C_d] / (1 + r_p)
C_0 = [(8/15) × 300 + (7/15) × 0] / 1.04 = 160 / 1.04 = **Rs 153.85**
*(Note: If annual compounding (1.08)^(180/360) = 1.03923 is used, p = 0.5316 and C_0 = **Rs 153.47**.)*

Part (b): Arbitrage Process and Percentage Return When Market Call Price (C_mkt) = Rs 140:
• Since Market Price (Rs 140) < Fair Value (Rs 153.85), the call option is **underpriced**!
• **Strategy:** Buy the underpriced Call Option and Short the Replicating Hedge Ratio (h) of Stock (Delta Hedging):
- **Hedge Ratio (h)** = (C_u - C_d) / (S_u - S_d) = (300 - 0) / (1,500 - 960) = 300 / 540 = **0.5556 shares per call (or 5 shares for every 9 calls)**.
• Portfolio Construction (Per 1 Share Shorted & 1/h = 1.8 Calls Bought, or Per 1 Call Bought & 0.5556 Share Shorted):
- Let us form the hedge portfolio with **1 Call Purchased** (cost = Rs 140) and **h = 5/9 share Shorted** (inflow = 5/9 × 1,200 = Rs 666.67):
Net Initial Cash Inflow (or Synthetic Borrowing) = Rs 666.67 - Rs 140.00 = **Rs 526.67**
- At expiration (both S_u = 1,500 and S_d = 960), the terminal obligation of the short stock minus long call is fixed at:
Terminal Obligation = (5/9) × 960 - 0 = **Rs 533.33** (or (5/9)×1,500 - 300 = 833.33 - 300 = Rs 533.33).
• Implied 6-Month Borrowing Rate vs. Risk-Free Rate & Arbitrage Return:
- Present value of Rs 533.33 at r_p = 4% is 533.33 / 1.04 = Rs 512.82, yielding an instant arbitrage profit today of Rs 153.85 - Rs 140 = **Rs 13.85 per call** (or Rs 13.85 × 1.04 = **Rs 14.40** at maturity).
- **Arbitrage Return (%) on Call Investment:** (153.85 - 140) / 140 = 13.85 / 140 = **9.89%** instant return (or on the hedged portfolio outlay of Rs 526.67: 6-month synthetic financing rate is (533.33 - 526.67) / 526.67 = **1.265%** for 6 months [**2.53% p.a.**] vs. investing at 4% for 6 months [8% p.a.], earning a net arbitrage spread of 4% - 1.265% = **2.735% for 6 months [5.47% p.a.]**).

EXAM HINT TO REMEMBER: *In Binomial Option Pricing, always compute Hedge Ratio h = (C_u - C_d)/(S_u - S_d) = 300/540 = 5/9 = 0.5556. If Market Call (140) < Fair Call (153.85), BUY Call and SHORT h shares of Stock!*

Q19. A one-year swap with quarterly payment and notional principal of Rs 100 million is entered between Citizen Investment Trust (CIT) and Global Bank. Payments will be based on a 30-day month and a 360-day

year. The CIT agreed to make interest payments based on the 90-day T-bill rate, and Global Bank agreed to make interest payments based on the 90-day SOFR minus a spread. a. Find the spread that should be deducted from the SOFR rate on each payment date. [Init Term Structure: 90d: SOFR 7.50%, T-bill 5.50%; 180d: SOFR 7.95%, T-bill 5.87%; 270d: SOFR 8.52%, T-bill 6.30%; 360d: SOFR 9.00%, T-bill 6.90%] b. Assume it is 60 days into the life of the swap. Find the value of the swap for both counterparties. [New Term Structure at 60d: 30d: SOFR 8.00%, T-bill 6.00%; 120d: SOFR 8.45%, T-bill 6.47%; 210d: SOFR 9.00%, T-bill 6.95%; 300d: SOFR 9.55%, T-bill 7.40%]

Answer (Step-by-Step Numerical Solution):

Given:

• Notional Principal (NP) = Rs 100,000,000 (Rs 100 million)

• Swap Type: 1-year quarterly Basis Swap (CIT pays 90-day T-bill rate; Global Bank pays 90-day SOFR minus a fixed spread).

Part (a): Calculation of the Spread to be Deducted from SOFR at Initiation (t = 0):

• **Step 1: Compute Discount Factors $B_i = 1 / [1 + R_i \times (Days_i / 360)]$ at Initiation:**

- **SOFR Discount Factors (B_SOFR):**

$B_1(90d) = 1 / [1 + 0.0750 \times (90/360)] = 1 / 1.01875 =$ **0.981595**

$B_2(180d) = 1 / [1 + 0.0795 \times (180/360)] = 1 / 1.03975 =$ **0.961770**

$B_3(270d) = 1 / [1 + 0.0852 \times (270/360)] = 1 / 1.06390 =$ **0.939938**

$B_4(360d) = 1 / [1 + 0.0900 \times (360/360)] = 1 / 1.09000 =$ **0.917431**

Sum of SOFR Discount Factors (Sum B_SOFR) = 0.981595 + 0.961770 + 0.939938 + 0.917431 = **3.800734**

- **T-Bill Discount Factors (B_Tbill):**

$B_1(90d) = 1 / [1 + 0.0550 \times (90/360)] = 1 / 1.01375 =$ **0.986436**

$B_2(180d) = 1 / [1 + 0.0587 \times (180/360)] = 1 / 1.02935 =$ **0.971487**

$B_3(270d) = 1 / [1 + 0.0630 \times (270/360)] = 1 / 1.04725 =$ **0.954882**

$B_4(360d) = 1 / [1 + 0.0690 \times (360/360)] = 1 / 1.06900 =$ **0.935454**

Sum of T-Bill Discount Factors (Sum B_Tbill) = 0.986436 + 0.971487 + 0.954882 + 0.935454 = **3.848259**

• **Step 2: Compute Fixed Swap Rates Equivalent to SOFR and T-Bill Legs:**

- Quarterly Fixed Rate for SOFR = (1 - B_4,SOFR) / Sum B_SOFR = (1 - 0.917431) / 3.800734 = 0.082569 / 3.800734 = **0.021724 (2.1724% per quarter)**

Annualized SOFR Swap Fixed Rate (R_SOFR) = 2.1724% × (360 / 90) = **8.6898% (or 8.69% p.a.)**

- Quarterly Fixed Rate for T-Bill = (1 - B_4,Tbill) / Sum B_Tbill = (1 - 0.935454) / 3.848259 = 0.064546 / 3.848259 = **0.016773 (1.6773% per quarter)**

Annualized T-Bill Swap Fixed Rate (R_Tbill) = 1.6773% × (360 / 90) = **6.7091% (or 6.71% p.a.)**

• **Step 3: Required Spread to Deduct from SOFR:**

- **Quarterly Spread** = 2.1724% - 1.6773% = **0.4952% (or 0.50% per quarter)**

- **Annualized Spread** = 8.6898% - 6.7091% = **1.9806% (or 1.98% p.a.)**

Part (b): Value of the Swap 60 Days into the Life of the Swap (Remaining Days: 30, 120, 210, 300):

• **Step 1: New Discount Factors at Day 60:**

- **New SOFR Discount Factors (B'_SOFR):**

$B'_1(30d) = 1 / [1 + 0.0800 \times (30/360)] =$ **0.993377**

$B'_2(120d) = 1 / [1 + 0.0845 \times (120/360)] =$ **0.972607**

$B'_3(210d) = 1 / [1 + 0.0900 \times (210/360)] =$ **0.950119**

$B'_4(300d) = 1 / [1 + 0.0955 \times (300/360)] =$ **0.926282**

Sum (Sum B'_SOFR) = **3.842385**

- **New T-Bill Discount Factors (B'_Tbill):**

$B'_1(30d) = 1 / [1 + 0.0600 \times (30/360)] =$ **0.995025**

$B'_2(120d) = 1 / [1 + 0.0647 \times (120/360)] =$ **0.978889**

$B'_3(210d) = 1 / [1 + 0.0695 \times (210/360)] =$ **0.961038**

$B'_4(300d) = 1 / [1 + 0.0740 \times (300/360)] =$ **0.941915**

Sum (Sum B'_Tbill) = **3.876867**

• **Step 2: Present Value of Both Legs at Day 60 (per Re 1 Notional):**

- First quarterly coupon (at Day 90, fixed at Day 0):

SOFR coupon = 0.0750 × (90/360) = 0.01875; T-Bill coupon = 0.0550 × (90/360) = 0.01375

- Value of Pure Floating SOFR Leg = (1 + 0.01875) × B'_1,SOFR = 1.01875 × 0.993377 = **1.012003**

- Value of Pure Floating T-Bill Leg = (1 + 0.01375) × B'_1,Tbill = 1.01375 × 0.995025 = **1.008706**

- Present Value of Spread Payments (0.004952 per quarter):

PV(Spread) = 0.004952 × Sum B'_SOFR = 0.004952 × 3.842385 = **0.019026**

*(Note: If only the net difference between pure floating legs is compared before spread PV: (1.012003 - 1.008706) × 100M = +Rs 329,700; or if next coupon includes spread 0.01875 - 0.004952 = 0.013798: V_SOFR-spread = 1.013798 × 0.993377 = 1.007084 -> Net Value to CIT = (1.007084 - 1.008706) × 100M = **-Rs 162,200**; or with full remaining spread stream: (1.012003 - 0.019026 - 1.008706) × 100M = **-Rs 1,572,902**.)*

***EXAM HINT TO REMEMBER:** In a Basis Swap (SOFR vs. T-Bill), first find the Fixed Rate for a SOFR swap and the Fixed Rate for a T-Bill swap using R = [(1 - B_4) / Sum(B)] × (360/90). The Spread is simply R_SOFR - R_Tbill = 8.69% - 6.71% = 1.98% p.a. (or 0.495% per quarter)!*

Q20. A stock is expected to pay a dividend of Rs 10 per share in two months and five months from today. The stock price is Rs 500, and the risk-free interest rate is 8% p.a. with continuous compounding for all maturities. An investor has just taken a short position in a six-month forward contract on the stock. a. What are the forward price and the initial value of the forward contract? b. Show the arbitrage process if the market price of the forward is Rs 515. c. Three months later, the stock price is Rs 480, and the risk-free rate of interest is still 8% p.a. What are the forward price and the value of the short position in the forward contract?

Answer (Step-by-Step Numerical Solution):

Given:

• Current stock price (S_0) = Rs 500

• Dividends: D_1 = Rs 10 in 2 months (t_1 = 2/12 yr); D_2 = Rs 10 in 5 months (t_2 = 5/12 yr)

• Risk-free interest rate (r) = 8% p.a. = 0.08 (continuously compounded)

• Maturity of forward contract (T) = 6 months = 6/12 = **0.5 year**

Part (a): Forward Price (F_0) and Initial Value of the Forward Contract:

• **Step 1: Present Value of Expected Dividends (I):**

I = D_1 × e^(-r × t_1) + D_2 × e^(-r × t_2)

I = 10 × e^(-0.08 × 2/12) + 10 × e^(-0.08 × 5/12)

I = 10 × e^(-0.013333) + 10 × e^(-0.033333)

I = 9.86755 + 9.67216 = **Rs 19.5397 (or Rs 19.54)**

• **Step 2: Fair 6-Month Forward Price (F_0):**

F_0 = (S_0 - I) × e^(r × T)

F_0 = (500 - 19.5397) × e^(0.08 × 0.5)

F_0 = 480.4603 × e^(0.04) = 480.4603 × 1.040811 = **Rs 500.07**

• **Step 3: Initial Value of the Forward Contract (V_0):**

At the time of initiation, the value of a forward contract is always **Rs 0**.

Part (b): Arbitrage Process If Market Forward Price (F_mkt) = Rs 515:

• Since F_mkt (Rs 515) > Fair F_0 (Rs 500.07), the forward contract is **overpriced** -> Execute **Cash-and-Carry Arbitrage** (Borrow, Buy Spot Stock, Short Forward):

- **Today (t = 0):**

1. Borrow Rs 500 at 8% continuous interest:

- Loan 1: Borrow Rs 9.8676 for 2 months (to be repaid using the first Rs 10 dividend at t = 2 months).

- Loan 2: Borrow Rs 9.6722 for 5 months (to be repaid using the second Rs 10 dividend at t = 5 months).

- Loan 3: Borrow the remaining Rs 480.4603 (= 500 - 19.5397) for 6 months.

2. Use the Rs 500 borrowed to buy 1 share of the stock in the spot market.

3. Enter into a **short** 6-month forward contract at F_mkt = Rs 515.

- **At 2 Months & 5 Months:** Collect each Rs 10 dividend and pay off Loan 1 and Loan 2 completely.

- **At Maturity (6 Months):**

1. Deliver the share under the short forward contract and receive **Rs 515.00**.

2. Repay Loan 3 with interest: Rs 480.4603 × e^(0.08 × 0.5) = **Rs 500.07**.

3. **Riskless Arbitrage Profit at Maturity** = Rs 515.00 - Rs 500.07 = **Rs 14.93 per share**.

Part (c): Forward Price and Value of Short Position Three Months Later (t = 3/12 = 0.25 yr):
• After 3 months, the first dividend (at 2 months) has already been paid! Only the second dividend of Rs 10 (at 5 months, which is now 5 - 3 = **2 months away**) remains, and remaining time to maturity is T - t = 6 - 3 = **3 months (0.25 year)**.
• Present value of remaining dividend at t = 3 months (I_t):
I_t = 10 × e^(-0.08 × 2/12) = 10 × e^(-0.013333) = **Rs 9.8676 (or Rs 9.87)**
• **New Forward Price at t = 3 months (F_t):**
F_t = (S_t - I_t) × e^[r × (T - t)]
F_t = (480 - 9.8676) × e^(0.08 × 3/12) = 470.1324 × e^(0.02) = 470.1324 × 1.020201 = **Rs 479.63**
• **Value of the Short Forward Position at t = 3 months (V_short):**
V_short = (F_0 - F_t) × e^[-r × (T - t)]
V_short = (500.07 - 479.63) × e^(-0.08 × 0.25) = 20.44 × 0.980199 = **+Rs 20.03 per share**
*(Alternative direct formula: V_short = F_0 × e^[-r(T-t)] - (S_t - I_t) = 500.07 × e^(-0.02) - (480 - 9.8676) = 490.168 - 470.132 = **+Rs 20.04**.)*

EXAM HINT TO REMEMBER: *Three months into a 6-month forward (Part c), watch out for two things: (1) The 2-month dividend is ALREADY gone; only the 5-month dividend remains, and it is now 5 - 3 = 2 months away! (2) Since the investor holds a SHORT position and the forward price dropped from Rs 500.07 to Rs 479.63, the short position has a POSITIVE value: (F_0 - F_t) × e^(-r(T-t)) = +Rs 20.03!*

Q21. Explain the margin rule and the margin operation process by a clearinghouse in option trading.

Answer:
1. Concept of Margin Rules in Option Trading:
In option trading, margin rules differ fundamentally between option **buyers (long positions)** and option **writers (short positions)** because their risk profiles are asymmetric:
• **Long Option Positions (Buyers of Calls and Puts):**
Generally, option buyers must pay 100% of the option premium upfront in cash. Once the premium is paid, the buyer has **no further margin requirement** because the buyer's maximum possible loss is limited to the premium already paid, and they have no future obligation to perform.
• **Short Option Positions (Writers of Naked or Covered Calls and Puts):**
An option writer receives the premium upfront but assumes a contingent liability to buy or sell the underlying asset if the option is exercised against them. Because potential losses on naked short calls are unlimited (and very large on short puts), exchanges and clearinghouses require option writers to post and maintain performance margin.
- *Naked (Uncovered) Option Writing Margin:* Typically calculated as the greater of two exchange formulas:
Formula 1: 100% of Option Proceeds + 20% of Underlying Stock Value - Out-of-the-Money Amount (if any).
Formula 2: 100% of Option Proceeds + 10% of Underlying Stock Value (for calls) or 10% of Exercise Price (for puts).
- *Covered Call Writing:* If the writer of a call option already owns and deposits the underlying shares in escrow with the broker, no separate cash margin is required on the short call because the shares guarantee delivery.

2. Margin Operation Process by a Clearinghouse in Option Trading:
• **Step 1: Novation by the Options Clearinghouse (e.g., OCC):** Once a trade is executed on the options exchange, the Clearinghouse steps in as the central counterparty—becoming the buyer to every option writer and the writer to every option buyer, eliminating bilateral credit risk.
• **Step 2: Initial Margin Deposit:** Option writers deposit Initial Margin with their brokerage firm (clearing member), which in turn posts clearing margin with the Clearinghouse in cash or Treasury securities.
• **Step 3: Daily Valuation and Marking-to-Market of Short Options:** At the end of each trading day, the Clearinghouse recalculates the required margin based on the closing stock price and closing option price.
• **Step 4: Maintenance Margin and Margin Calls:** If the underlying stock moves against the option writer (e.g., stock price rises sharply against a naked call writer), the required maintenance margin increases. If the writer's account equity falls below the maintenance requirement, the broker issues a **margin call** requiring immediate cash deposit.
• **Step 5: Forced Liquidation or Exercise Settlement:** If the writer fails to meet the margin call, the broker closes out the short option position by buying back the option in the market. When an option holder chooses to exercise, the Clearinghouse randomly assigns the exercise notice to a clearing member with a short position, using the margin/collateral to guarantee settlement.

Group - D: Comprehensive-Answer / Case / Situation Analysis [20 Marks]

Q22. Case Study: Krishna Pauroti Company Stock closed at S_0 = Rs 240 today. Call and Put options with Exercise Price X = Rs 240 and T = 3 months to expiration closed at C = Rs 40 and P = Rs 20, respectively.

Risk-free rate r = 6% p.a. Total investment budget = Rs 240,000. a. If you believe the court verdict will favor the company and want to speculate on the option, which option do you buy? Why? How much will you earn/lose if the stock price after three months increases to Rs 350? What will be your annual rate of return? b. If you believe that the court verdict will come against the company and want to speculate on the option, which option do you buy? Why? How much will you earn/lose if the stock price after three months decreases to Rs 250 [also solved for Rs 150]? What will be your annual rate of return? Compare the risk and return of your investment with that of your stock investment. c. Assume you cannot guess whether the court verdict may come in favor of or against the company. Is a long straddle beneficial in such conditions? Why? Calculate gain/loss from long straddle for expiration stock prices of Rs 160, Rs 180, … Rs 320. After three months, calculate the maximum possible gain, maximum possible loss, and breakeven stock price(s) for your strategy. d. Considering the uncertainty of the verdict, you want to ensure your portfolio of stocks using put options. How many stocks and put options do you need to hold? What will be the ensured value of your portfolio? What will be the value of the ensured portfolio if the verdict came in favor of the company, and the stock price increased to Rs 350? Also, calculate the cost of insurance and the upside capture. [4+4+6+6]

Answer (Step-by-Step Comprehensive Case Solution):

Given Summary:

• Current Stock Price (S_0) = Rs 240; Exercise Price (X) = Rs 240; Time (T) = 3 months (0.25 year)

• Call Premium (C) = Rs 40; Put Premium (P) = Rs 20; Risk-free rate (r) = 6% p.a.

• Total Capital to Invest = **Rs 240,000**

Part (a): Bullish Speculation (Court Verdict Favors the Company) [4 Marks]:

• **Which Option to Buy & Why:** You should **buy Call Options** on Krishna stock. If the court verdict favors the company, the stay order will be lifted, capacity expansion will proceed, and the stock price will surge above Rs 240. Call options give high leveraged upside participation with limited downside risk.

• **Number of Call Options Purchased:** Rs 240,000 / Rs 40 = **6,000 call options**.

• **Total Dollar Gain/Loss if S_T = Rs 350 after 3 Months:**

- Profit per call = max(S_T - X, 0) - C = (350 - 240) - 40 = 110 - 40 = **Rs 70 per call**

- **Total Net Profit** = 6,000 calls × Rs 70 = **+Rs 420,000**

• **Annual Rate of Return:**

- 3-Month Holding Period Return (HPR) = Rs 420,000 / Rs 240,000 = 70 / 40 = **175%**

- **Simple Annualized Rate of Return** = 175% × (12 / 3) = **700% per annum**

*(Compound Effective Annual Return = (1 + 1.75)^4 - 1 = **5,618.75% p.a.**)*

Part (b): Bearish Speculation (Court Verdict Comes Against the Company) [4 Marks]:

• **Which Option to Buy & Why:** You should **buy Put Options** on Krishna stock because an adverse verdict will permanently ban the popular bread brand, causing the stock price to plunge below the strike price of Rs 240.

• **Number of Put Options Purchased:** Rs 240,000 / Rs 20 = **12,000 put options**.

• **Gain/Loss and Annual Return After 3 Months:**

- **Case 1 (At literally printed S_T = Rs 250):**

Since S_T = Rs 250 > Strike X = Rs 240, the put option is out-of-the-money and expires worthless!

- Loss per put = -Rs 20 -> **Total Loss = -Rs 240,000** (100% of investment lost).

- 3-Month HPR = **-100%**; Simple Annualized Return = -100% × (12/3) = **-400% p.a.**

- Meanwhile, an investor who bought 1,000 shares of stock at Rs 240 would have Rs 250,000 (+Rs 10,000 profit; +4.17% in 3 months or +16.67% p.a.).

- **Case 2 (If "decreases to Rs 250" in the question paper is a misprint for "decreases to Rs 150"):**

At S_T = Rs 150:

- Profit per put = max(240 - 150, 0) - 20 = 90 - 20 = **+Rs 70 per put**

- **Total Net Profit** = 12,000 puts × Rs 70 = **+Rs 840,000**

- 3-Month HPR = 840,000 / 240,000 = **+350%**; Simple Annualized Return = 350% × 4 = **+1,400% p.a.**

- Meanwhile, buying 1,000 shares of stock at Rs 240 and holding to Rs 150 results in a loss of (150 - 240) × 1,000 = **-Rs 90,000** (-37.5% in 3 months or -150% p.a.).

• **Risk & Return Comparison (Options vs. Stock):** Options provide extreme financial leverage—magnifying positive percentage returns when the forecast is right, but carrying a 100% loss risk of the entire Rs 240,000 capital if the option finishes out-of-the-money, whereas direct stock ownership has much lower volatility and only loses 100% if the company's share price drops to zero.

Part (c): Long Straddle Strategy Under Verdict Uncertainty [6 Marks]:

• Is a Long Straddle Beneficial & Why?

Yes, a Long Straddle is highly beneficial. A long straddle involves buying 1 Call (X = 240, C = 40) and 1 Put (X = 240, P = 20) simultaneously at a combined cost of C + P = 40 + 20 = **Rs 60 per straddle**. Because the court verdict is a binary event that will either send the stock soaring (if favorable) or crashing (if unfavorable), volatility will be huge, allowing one leg to generate large profits that more than offset the Rs 60 combined premium.

• Number of Straddles with Rs 240,000: Rs 240,000 / Rs 60 = **4,000 straddles** (4,000 calls + 4,000 puts).

• Maximum Possible Gain, Maximum Possible Loss, and Breakeven Stock Prices:

- **Maximum Possible Gain: Unlimited** on the upside (as S_T -> Infinity, profit -> Infinity); on the downside (if S_T falls to Rs 0), max downside gain is Rs 240 - Rs 60 = **Rs 180 per straddle** (or 4,000 × 180 = **Rs 720,000**).
- **Maximum Possible Loss: Rs 60 per straddle** (or **Rs 240,000 total**), occurring only if S_T = Rs 240 at expiration.
- **Two Breakeven Stock Prices (BEP):**

1. Lower Breakeven Price = X - (C + P) = 240 - 60 = **Rs 180**
2. Upper Breakeven Price = X + (C + P) = 240 + 60 = **Rs 300**

Expiration Stock Price (S_T)	Call Payoff max(S_T-240, 0)	Put Payoff max(240-S_T, 0)	Combined Payoff (Rs)	Initial Cost C + P (Rs)	Net Gain/Loss per Straddle (Rs)	Total Gain/Loss (4,000 Straddles)
Rs 160	0	80	80	60	**+Rs 20**	**+Rs 80,000**
Rs 180 (Lower BEP)	0	60	60	60	**Rs 0**	**Rs 0**
Rs 200	0	40	40	60	-Rs 20	-Rs 80,000
Rs 220	0	20	20	60	-Rs 40	-Rs 160,000
Rs 240 (Max Loss)	0	0	0	60	**-Rs 60**	**-Rs 240,000**
Rs 260	20	0	20	60	-Rs 40	-Rs 160,000
Rs 280	40	0	40	60	-Rs 20	-Rs 80,000
Rs 300 (Upper BEP)	60	0	60	60	**Rs 0**	**Rs 0**
Rs 320	80	0	80	60	**+Rs 20**	**+Rs 80,000**

Part (d): Portfolio Insurance Using Protective Put [6 Marks]:

• 1. Number of Stocks and Put Options to Hold (Protective Put Strategy):

- Cost of 1 Insured Unit (1 Share of Stock @ Rs 240 + 1 Put Option @ Rs 20) = Rs 240 + Rs 20 = **Rs 260 per unit**.
- With a fixed budget of Rs 240,000:

Number of Shares and Put Options = Rs 240,000 / Rs 260 = **923.08 shares and 923.08 put options** (or **923 shares and 923 puts** with Rs 20 cash left over).

(Note: If an investor already holds 1,000 shares worth Rs 240,000 and buys 1,000 puts for an extra Rs 20,000 [total outlay Rs 260,000], the corresponding figures are also shown below!)

• 2. Ensured (Floor) Value of the Portfolio:

- Regardless of how low the stock price falls, each share can be sold at the put strike price X = Rs 240:

Ensured Floor Value = 923.0769 × Rs 240 = **Rs 221,538.46** (or 923 × Rs 240 = **Rs 221,520**; and for 1,000 shares = **Rs 240,000**).

• 3. Value of the Ensured Portfolio if S_T Increases to Rs 350:

- At S_T = Rs 350, the put options expire worthless (0) and each share is worth Rs 350:

Value of Ensured Portfolio at S_T = Rs 350 = 923.0769 × Rs 350 = **Rs 323,076.92** (or 923 × Rs 350 = **Rs 323,050**; and for 1,000 shares, net value after Rs 20,000 put cost = Rs 350,000 - Rs 20,000 = **Rs 330,000**).

• 4. Cost of Insurance and Upside Capture:

- **Uninsured Stock Portfolio Value at S_T = Rs 350:** Investing Rs 240,000 solely in stock buys 1,000 shares (240,000 / 240), which grows to 1,000 × Rs 350 = **Rs 350,000** (Gain = Rs 110,000 or +45.83%).
- **Cost of Insurance:**

• In total Put Premium paid (for 923.08 units) = 923.0769 × Rs 20 = **Rs 18,461.54** (or **Rs 20 per share** / **Rs 20,000** for 1,000 shares, which is 20/260 = **7.69%** of portfolio value).

• In terms of foregone upside terminal wealth at S_T = 350: Rs 350,000 - Rs 323,076.92 = **Rs 26,923.08**.

- **Upside Capture:**
• **Terminal Wealth Upside Capture Ratio** = (Insured Portfolio Terminal Value / Uninsured Portfolio Terminal Value) × 100 = (Rs 323,076.92 / Rs 350,000) × 100 = **92.31%** (or on 1,000 shares net of premium: 330,000 / 350,000 = **94.29%**).
• **Capital Gain / Return Upside Capture Ratio** = (Insured Net Gain / Uninsured Net Gain) × 100 = (Rs 83,076.92 / Rs 110,000) × 100 = **75.52%** (or on 1,000 shares: (330,000 - 240,000) / 110,000 = 90,000 / 110,000 = **81.82%**).

EXAM HINT TO REMEMBER: *In Protective Put Portfolio Insurance (Q22d) with Rs 240,000 budget: Cost per insured share = S_0 + P = 240 + 20 = Rs 260, so Shares = Puts = 240,000 / 260 = 923.08! Ensured Floor = 923.08 × 240 = Rs 221,538.46, and Terminal Value Upside Capture = S_0 / (S_0 + P) = 240 / 260 = 92.31%!*